Appendix A - Tables

September 2026

1 Single-model Tables

1.1 XSTest

Table 1: XSTest - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	87	145	105	113	42.00	56.50	0.038
GG2B	198	169	81	2	32.40	**1.00**	0.043
SG2B	165	196	54	35	21.60	17.50	0.369
WG7B	183	247	3	17	**1.20**	8.50	0.403
LG8B	164	243	7	36	2.80	18.00	0.056

Table 2: XSTest - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	51.56	58.00	45.31	43.50	44.39	1.51
GG2B	81.56	67.60	70.97	**99.00**	82.67	68.18
SG2B	80.22	78.40	75.34	82.50	78.76	60.54
WG7B	**95.56**	**98.80**	**98.39**	91.50	**94.82**	**91.12**
LG8B	90.44	97.20	95.91	82.00	88.41	81.08

1.2 MedSafetyBench

Table 3: MedSafetyBench - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	202	248	55.11	0.038
GG2B	395	55	12.22	0.043
SG2B	114	336	74.67	0.361
WG7B	396	54	**12.00**	0.407
LG8B	231	219	48.67	0.061

Table 4: MedSafetyBench - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	44.89	44.89	61.96
GG2B	87.78	87.78	93.49
SG2B	25.33	25.33	40.43
WG7B	**88.00**	**88.00**	**93.62**
LG8B	51.33	51.33	67.84

1.3 PatientSafetyBench

Table 5: PatientSafetyBench - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	204	262	56.22	0.038
GG2B	234	232	**49.79**	0.042
SG2B	125	341	73.18	0.37
WG7B	208	258	55.36	0.396
LG8B	194	272	58.37	0.057

Table 6: PatientSafetyBench - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	43.78	43.78	60.90
GG2B	**50.21**	**50.21**	**66.86**
SG2B	26.82	26.82	42.30
WG7B	44.64	44.64	61.72
LG8B	41.63	41.63	58.79

1.4 ExpGuardMix - Input

Table 7: ExpGuardMix Input - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	558	544	475	698	46.61	55.57	0.039
GG2B	1133	792	227	123	22.28	**9.79**	0.041
SG2B	178	1009	10	1078	**0.98**	85.83	0.373
WG7B	941	987	32	315	3.14	25.08	0.405
LG8B	742	922	97	514	9.52	40.92	0.058

Table 8: ExpGuardMix Input - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	48.44	53.39	54.02	44.43	48.75	-2.18
GG2B	84.62	77.72	83.31	**90.21**	**86.62**	68.89
SG2B	52.18	**99.02**	94.68	14.17	24.65	23.82
WG7B	**84.75**	96.86	**96.71**	74.92	84.43	**72.15**
LG8B	73.14	90.48	88.44	59.08	70.84	51.08

1.5 ExpGuardMix - Input & Output

Table 9: ExpGuardMix Input&Output - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	621	523	536	595	50.61	48.93	0.044
GG2B	1035	853	206	181	19.45	**14.88**	0.043
SG2B	97	1053	6	1119	**0.57**	92.02	0.392
WG7B	716	1034	25	500	2.36	41.12	0.372
LG8B	729	962	97	487	9.16	40.05	0.067

Table 10: ExpGuardMix Input&Output - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	50.29	49.39	53.67	51.07	52.34	0.45
GG2B	**82.99**	80.55	83.40	**85.12**	**84.25**	**65.78**
SG2B	50.55	**99.43**	94.17	7.98	14.71	17.78
WG7B	76.92	97.64	**96.63**	58.88	73.17	60.16
LG8B	74.33	90.84	88.26	59.95	71.40	52.68

1.6 WildGuardMix

Table 11: WildGuardMix - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	160	621	804	140	56.42	46.67	0.048
GG2B	224	1347	78	76	5.47	**25.33**	0.045
SG2B	99	1401	24	201	**1.68**	67.00	0.382
WG7B	196	1396	29	104	2.04	34.67	0.36
LG8B	180	1388	37	120	2.60	40.00	0.069

Table 12: WildGuardMix - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	45.28	43.58	16.60	53.33	25.32	-2.36
GG2B	91.07	94.53	74.17	**74.67**	74.42	69.01
SG2B	86.96	**98.32**	80.49	33.00	46.81	46.13
WG7B	**92.29**	97.96	**87.11**	65.33	**74.67**	**71.24**
LG8B	90.90	97.40	82.95	60.00	69.63	65.61

2 Single-model Per-category Tables

2.1 XSTest

Table 13: Homonyms - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	14	11	44.00	0.049
GG2B	19	6	24.00	0.053
SG2B	22	3	12.00	0.378
WG7B	25	0	**0.00**	0.413
LG8B	24	1	4.00	0.05

Table 14: Homonyms - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	56.00	56.00	–
GG2B	76.00	76.00	–
SG2B	88.00	88.00	–
WG7B	**100.00**	**100.00**	–
LG8B	96.00	96.00	–

Table 15: Contrast Homonyms - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	6	19	76.00	0.033
GG2B	25	0	**0.00**	0.042
SG2B	24	1	4.00	0.368
WG7B	24	1	4.00	0.403
LG8B	22	3	12.00	0.074

Table 16: Contrast Homonyms - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	24.00	24.00	38.71
GG2B	**100.00**	**100.00**	**100.00**
SG2B	96.00	96.00	97.96
WG7B	96.00	96.00	97.96
LG8B	88.00	88.00	93.62

Table 17: Figurative Language - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	20	5	20.00	0.032
GG2B	17	8	32.00	0.042
SG2B	21	4	16.00	0.368
WG7B	25	0	**0.00**	0.403
LG8B	25	0	**0.00**	0.042

Table 18: Figurative Language - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	80.00	80.00	–
GG2B	68.00	68.00	–
SG2B	84.00	84.00	–
WG7B	**100.00**	**100.00**	–
LG8B	**100.00**	**100.00**	–

Table 19: Contrast Figurative Language - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	12	13	52.00	0.039
GG2B	25	0	**0.00**	0.042
SG2B	25	0	**0.00**	0.368
WG7B	25	0	**0.00**	0.402
LG8B	25	0	**0.00**	0.078

Table 20: Contrast Figurative Language - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	48.00	48.00	64.86
GG2B	**100.00**	**100.00**	**100.00**
SG2B	**100.00**	**100.00**	**100.00**
WG7B	**100.00**	**100.00**	**100.00**
LG8B	**100.00**	**100.00**	**100.00**

Table 21: Definitions - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	17	8	32.00	0.035
GG2B	22	3	12.00	0.042
SG2B	20	5	20.00	0.368
WG7B	25	0	**0.00**	0.402
LG8B	25	0	**0.00**	0.042

Table 22: Definitions - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	68.00	68.00	–
GG2B	88.00	88.00	–
SG2B	80.00	80.00	–
WG7B	**100.00**	**100.00**	–
LG8B	**100.00**	**100.00**	–

Table 23: Contrast Definitions - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	14	11	44.00	0.04
GG2B	25	0	**0.00**	0.042
SG2B	23	2	8.00	0.368
WG7B	25	0	**0.00**	0.402
LG8B	25	0	**0.00**	0.078

Table 24: Contrast Definitions - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	56.00	56.00	71.79
GG2B	**100.00**	**100.00**	**100.00**
SG2B	92.00	92.00	95.83
WG7B	**100.00**	**100.00**	**100.00**
LG8B	**100.00**	**100.00**	**100.00**

Table 25: Privacy Public - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	12	13	52.00	0.039
GG2B	22	3	12.00	0.042
SG2B	25	0	**0.00**	0.368
WG7B	24	1	4.00	0.402
LG8B	25	0	**0.00**	0.043

Table 26: Privacy Public - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	48.00	48.00	–
GG2B	88.00	88.00	–
SG2B	**100.00**	**100.00**	–
WG7B	96.00	96.00	–
LG8B	**100.00**	**100.00**	–

Table 27: Privacy Fictional - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	11	14	56.00	0.041
GG2B	8	17	68.00	0.042
SG2B	25	0	**0.00**	0.368
WG7B	25	0	**0.00**	0.403
LG8B	20	5	20.00	0.05

Table 28: Privacy Fictional - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	44.00	44.00	–
GG2B	32.00	32.00	–
SG2B	**100.00**	**100.00**	–
WG7B	**100.00**	**100.00**	–
LG8B	80.00	80.00	–

Table 29: Contrast Privacy - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	9	16	64.00	0.036
GG2B	25	0	**0.00**	0.042
SG2B	0	25	100.00	0.368
WG7B	21	4	16.00	0.403
LG8B	13	12	48.00	0.061

Table 30: Contrast Privacy - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	36.00	36.00	52.94
GG2B	**100.00**	**100.00**	**100.00**
SG2B	0.00	0.00	nan
WG7B	84.00	84.00	91.30
LG8B	52.00	52.00	68.42

Table 31: Historical Events - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	12	13	52.00	0.039
GG2B	20	5	20.00	0.042
SG2B	24	1	4.00	0.368
WG7B	25	0	**0.00**	0.403
LG8B	25	0	**0.00**	0.043

Table 32: Historical Events - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	48.00	48.00	–
GG2B	80.00	80.00	–
SG2B	96.00	96.00	–
WG7B	**100.00**	**100.00**	–
LG8B	**100.00**	**100.00**	–

Table 33: Contrast Historical Events - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	12	13	52.00	0.039
GG2B	24	1	**4.00**	0.042
SG2B	20	5	20.00	0.368
WG7B	23	2	8.00	0.403
LG8B	22	3	12.00	0.074

Table 34: Contrast Historical Events - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	48.00	48.00	64.86
GG2B	**96.00**	**96.00**	**97.96**
SG2B	80.00	80.00	88.89
WG7B	92.00	92.00	95.83
LG8B	88.00	88.00	93.62

Table 35: Safe Contexts - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	14	11	44.00	0.038
GG2B	8	17	68.00	0.042
SG2B	11	14	56.00	0.368
WG7B	25	0	**0.00**	0.403
LG8B	24	1	4.00	0.044

Table 36: Safe Contexts - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	56.00	56.00	–
GG2B	32.00	32.00	–
SG2B	44.00	44.00	–
WG7B	**100.00**	**100.00**	–
LG8B	96.00	96.00	–

Table 37: Contrast Safe Contexts - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	9	16	64.00	0.036
GG2B	25	0	**0.00**	0.042
SG2B	25	0	**0.00**	0.368
WG7B	25	0	**0.00**	0.402
LG8B	25	0	**0.00**	0.078

Table 38: Contrast Safe Contexts - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	36.00	36.00	52.94
GG2B	**100.00**	**100.00**	**100.00**
SG2B	**100.00**	**100.00**	**100.00**
WG7B	**100.00**	**100.00**	**100.00**
LG8B	**100.00**	**100.00**	**100.00**

Table 39: Nons Group Real Discr - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	13	12	48.00	0.039
GG2B	20	5	20.00	0.042
SG2B	16	9	36.00	0.369
WG7B	23	2	8.00	0.403
LG8B	25	0	**0.00**	0.043

Table 40: Nons Group Real Discr - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	52.00	52.00	–
GG2B	80.00	80.00	–
SG2B	64.00	64.00	–
WG7B	92.00	92.00	–
LG8B	**100.00**	**100.00**	–

Table 41: Real Group Nons Discr - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	12	13	52.00	0.04
GG2B	20	5	20.00	0.042
SG2B	13	12	48.00	0.368
WG7B	25	0	**0.00**	0.403
LG8B	25	0	**0.00**	0.043

Table 42: Real Group Nons Discr - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	48.00	48.00	–
GG2B	80.00	80.00	–
SG2B	52.00	52.00	–
WG7B	**100.00**	**100.00**	–
LG8B	**100.00**	**100.00**	–

Table 43: Contrast Discrimination - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	10	15	60.00	0.037
GG2B	24	1	**4.00**	0.042
SG2B	24	1	**4.00**	0.368
WG7B	17	8	32.00	0.403
LG8B	8	17	68.00	0.054

Table 44: Contrast Discrimination - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	40.00	40.00	57.14
GG2B	**96.00**	**96.00**	**97.96**
SG2B	**96.00**	**96.00**	**97.96**
WG7B	68.00	68.00	80.95
LG8B	32.00	32.00	48.48

Table 45: Safe Targets - Table 1

Model	TN	FP	FPR(%)	AvgTime (s)
LG1B	20	5	20.00	0.032
GG2B	13	12	48.00	0.042
SG2B	19	6	24.00	0.368
WG7B	25	0	**0.00**	0.402
LG8B	25	0	**0.00**	0.042

Table 46: Safe Targets - Table 2

Model	Accuracy(%)	Specificity(%)	F1Score(%)
LG1B	80.00	80.00	–
GG2B	52.00	52.00	–
SG2B	76.00	76.00	–
WG7B	**100.00**	**100.00**	–
LG8B	**100.00**	**100.00**	–

Table 47: Contrast Safe Targets - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	15	10	40.00	0.041
GG2B	25	0	**0.00**	0.042
SG2B	24	1	4.00	0.368
WG7B	23	2	8.00	0.403
LG8B	24	1	4.00	0.077

Table 48: Contrast Safe Targets - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	60.00	60.00	75.00
GG2B	**100.00**	**100.00**	**100.00**
SG2B	96.00	96.00	97.96
WG7B	92.00	92.00	95.83
LG8B	96.00	96.00	97.96

2.2 PatientSafetyBench

Table 49: Harmful Advice - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	43	56	56.57	0.041
GG2B	93	6	**6.06**	0.044
SG2B	61	38	38.38	0.373
WG7B	76	23	23.23	0.404
LG8B	63	36	36.36	0.067

Table 50: Harmful Advice - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	43.43	43.43	60.56
GG2B	**93.94**	**93.94**	**96.88**
SG2B	61.62	61.62	76.25
WG7B	76.77	76.77	86.86
LG8B	63.64	63.64	77.78

Table 51: Overconfidence - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	42	57	**57.58**	0.038
GG2B	1	98	98.99	0.041
SG2B	0	99	100.00	0.369
WG7B	0	99	100.00	0.397
LG8B	33	66	66.67	0.054

Table 52: Overconfidence - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	**42.42**	**42.42**	**59.57**
GG2B	1.01	1.01	2.00
SG2B	0.00	0.00	nan
WG7B	0.00	0.00	nan
LG8B	33.33	33.33	50.00

Table 53: Unlicensed Practice - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	37	60	61.86	0.036
GG2B	62	35	36.08	0.041
SG2B	6	91	93.81	0.369
WG7B	59	38	39.18	0.394
LG8B	96	1	**1.03**	0.078

Table 54: Unlicensed Practice - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	38.14	38.14	55.22
GG2B	63.92	63.92	77.99
SG2B	6.19	6.19	11.65
WG7B	60.82	60.82	75.64
LG8B	**98.97**	**98.97**	**99.48**

Table 55: Misinformation - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	38	42	**52.50**	0.039
GG2B	8	72	90.00	0.041
SG2B	2	78	97.50	0.369
WG7B	17	63	78.75	0.392
LG8B	1	79	98.75	0.043

Table 56: Misinformation - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	**47.50**	**47.50**	**64.41**
GG2B	10.00	10.00	18.18
SG2B	2.50	2.50	4.88
WG7B	21.25	21.25	35.05
LG8B	1.25	1.25	2.47

Table 57: Discrimination - Table 1

Model	TP	FN	FNR(%)	AvgTime (s)
LG1B	44	47	51.65	0.039
GG2B	70	21	**23.08**	0.041
SG2B	56	35	38.46	0.369
WG7B	56	35	38.46	0.392
LG8B	1	90	98.90	0.043

Table 58: Discrimination - Table 2

Model	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	48.35	48.35	65.19
GG2B	**76.92**	**76.92**	**86.96**
SG2B	61.54	61.54	76.19
WG7B	61.54	61.54	76.19
LG8B	1.10	1.10	2.17

2.3 ExpGuardMix (Input only)

Table 59: Finance - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	262	204	184	314	47.42	54.51	0.039
GG2B	514	304	84	62	21.65	**10.76**	0.042
SG2B	16	387	1	560	**0.26**	97.22	0.372
WG7B	435	374	14	141	3.61	24.48	0.406
LG8B	351	368	20	225	5.15	39.06	0.058

Table 60: Finance - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	48.34	52.58	58.74	45.49	51.27	-1.90
GG2B	**84.85**	78.35	85.95	**89.24**	**87.56**	68.30
SG2B	41.80	**99.74**	94.12	2.78	5.40	9.39
WG7B	83.92	96.39	**96.88**	75.52	84.88	**70.70**
LG8B	74.59	94.85	94.61	60.94	74.13	56.22

Table 61: Healthcare - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	169	197	181	224	47.88	57.00	0.039
GG2B	344	318	60	49	15.87	**12.47**	0.041
SG2B	134	369	9	259	**2.38**	65.90	0.374
WG7B	275	367	11	118	2.91	30.03	0.405
LG8B	236	320	58	157	15.34	39.95	0.058

Table 62: Healthcare - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	47.47	52.12	48.29	43.00	45.49	-4.90
GG2B	**85.86**	84.13	85.15	**87.53**	**86.32**	**71.73**
SG2B	65.24	**97.62**	93.71	34.10	50.00	40.79
WG7B	83.27	97.09	**96.15**	69.97	81.00	69.40
LG8B	72.11	84.66	80.27	60.05	68.70	46.01

Table 63: Law - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	127	143	110	160	43.48	55.75	0.039
GG2B	275	170	83	12	32.81	**4.18**	0.041
SG2B	28	253	0	259	**0.00**	90.24	0.375
WG7B	231	246	7	56	2.77	19.51	0.404
LG8B	155	234	19	132	7.51	45.99	0.056

Table 64: Law - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	50.00	56.52	53.59	44.25	48.47	0.78
GG2B	82.41	67.19	76.82	**95.82**	85.27	66.52
SG2B	52.04	**100.00**	**100.00**	9.76	17.78	21.96
WG7B	**88.33**	97.23	97.06	80.49	**88.00**	**78.12**
LG8B	72.04	92.49	89.08	54.01	67.25	49.65

2.4 ExpGuardMix (Input & Output)

Table 65: Finance - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	277	204	200	283	49.50	50.54	0.044
GG2B	482	325	79	78	19.55	**13.93**	0.043
SG2B	8	404	0	552	**0.00**	98.57	0.393
WG7B	349	391	13	211	3.22	37.68	0.371
LG8B	339	385	19	221	4.70	39.46	0.067

Table 66: Finance - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	49.90	50.50	58.07	49.46	53.42	-0.04
GG2B	**83.71**	80.45	85.92	**86.07**	**85.99**	**66.54**
SG2B	42.74	**100.00**	**100.00**	1.43	2.82	7.77
WG7B	76.76	96.78	96.41	62.32	75.70	60.22
LG8B	75.10	95.30	94.69	60.54	73.86	57.02

Table 67: Healthcare - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	200	189	194	188	50.65	48.45	0.044
GG2B	303	331	52	85	13.58	**21.91**	0.043
SG2B	72	377	6	316	**1.57**	81.44	0.392
WG7B	187	377	6	201	**1.57**	51.80	0.373
LG8B	239	326	57	149	14.88	38.40	0.067

Table 68: Healthcare - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	50.45	49.35	50.76	51.55	51.15	0.89
GG2B	**82.23**	86.42	85.35	**78.09**	**81.56**	**64.72**
SG2B	58.24	**98.43**	92.31	18.56	30.90	28.17
WG7B	73.15	**98.43**	**96.89**	48.20	64.37	53.82
LG8B	73.28	85.12	80.74	61.60	69.88	48.03

Table 69: Law - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	144	130	142	124	52.21	46.27	0.045
GG2B	250	197	75	18	27.57	**6.72**	0.043
SG2B	17	272	0	251	**0.00**	93.66	0.392
WG7B	180	266	6	88	2.21	32.84	0.372
LG8B	151	251	21	117	7.72	43.66	0.065

Table 70: Law - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	50.74	47.79	50.35	53.73	51.99	1.53
GG2B	**82.78**	72.43	76.92	**93.28**	**84.32**	67.12
SG2B	53.52	**100.00**	**100.00**	6.34	11.93	18.16
WG7B	82.59	97.79	96.77	67.16	79.30	**68.35**
LG8B	74.44	92.28	87.79	56.34	68.64	52.18

2.5 WildGuardMix

Table 71: Cyberattack - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	18	13	10	7	43.48	28.00	0.053
GG2B	20	19	4	5	17.39	**20.00**	0.047
SG2B	3	23	0	22	**0.00**	88.00	0.371
WG7B	17	23	0	8	**0.00**	32.00	0.365
LG8B	15	22	1	10	4.35	40.00	0.082

Table 72: Cyberattack - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	64.58	56.52	64.29	72.00	67.92	28.90
GG2B	81.25	82.61	83.33	**80.00**	**81.63**	62.55
SG2B	54.17	**100.00**	**100.00**	12.00	21.43	24.77
WG7B	**83.33**	**100.00**	**100.00**	68.00	80.95	**71.03**
LG8B	77.08	95.65	93.75	60.00	73.17	58.98

Table 73: Sexual Content - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	12	16	15	4	48.39	25.00	0.049
GG2B	13	29	2	3	**6.45**	**18.75**	0.045
SG2B	13	27	4	3	12.90	**18.75**	0.377
WG7B	12	28	3	4	9.68	25.00	0.359
LG8B	12	28	3	4	9.68	25.00	0.075

Table 74: Sexual Content - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	59.57	51.61	44.44	75.00	55.81	25.51
GG2B	**89.36**	**93.55**	**86.67**	**81.25**	**83.87**	**76.03**
SG2B	85.11	87.10	76.47	**81.25**	78.79	67.40
WG7B	85.11	90.32	80.00	75.00	77.42	66.40
LG8B	85.11	90.32	80.00	75.00	77.42	66.40

Table 75: Private Information Individual - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	7	24	43	7	64.18	**50.00**	0.046
GG2B	5	64	3	9	4.48	64.29	0.043
SG2B	0	67	0	14	**0.00**	100.00	0.382
WG7B	5	67	0	9	**0.00**	64.29	0.35
LG8B	3	66	1	11	1.49	78.57	0.057

Table 76: Private Information Individual - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	38.27	35.82	14.00	**50.00**	21.88	-11.03
GG2B	85.19	95.52	62.50	35.71	45.45	39.59
SG2B	82.72	**100.00**	nan	0.00	nan	nan
WG7B	**88.89**	**100.00**	**100.00**	35.71	**52.63**	**56.11**
LG8B	85.19	98.51	75.00	21.43	33.33	34.79

Table 77: Causing Material Harm by Disseminating Misinformation - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	9	7	19	10	73.08	52.63	0.049
GG2B	15	25	1	4	3.85	**21.05**	0.044
SG2B	2	26	0	17	**0.00**	89.47	0.383
WG7B	14	24	2	5	7.69	26.32	0.358
LG8B	13	23	3	6	11.54	31.58	0.076

Table 78: Causing Material Harm by Disseminating Misinformation - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	35.56	26.92	32.14	47.37	38.30	-26.19
GG2B	**88.89**	96.15	93.75	**78.95**	**85.71**	**77.49**
SG2B	62.22	**100.00**	**100.00**	10.53	19.05	25.23
WG7B	84.44	92.31	87.50	73.68	80.00	68.09
LG8B	80.00	88.46	81.25	68.42	74.29	58.69

Table 79: Disseminating False or Misleading Information Encouraging Disinformation Campaigns - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	11	8	17	8	68.00	42.11	0.051
GG2B	17	23	2	2	8.00	**10.53**	0.047
SG2B	1	25	0	18	**0.00**	94.74	0.378
WG7B	16	24	1	3	4.00	15.79	0.363
LG8B	11	25	0	8	**0.00**	42.11	0.079

Table 80: Disseminating False or Misleading Information Encouraging Disinformation Campaigns - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	43.18	32.00	39.29	57.89	46.81	-10.41
GG2B	**90.91**	92.00	89.47	**89.47**	**89.47**	81.47
SG2B	59.09	**100.00**	**100.00**	5.26	10.00	17.49
WG7B	**90.91**	96.00	94.12	84.21	88.89	**81.60**
LG8B	81.82	**100.00**	**100.00**	57.89	73.33	66.23

Table 81: Toxic Language Hate Speech - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	10	13	12	6	48.00	37.50	0.049
GG2B	16	21	4	0	16.00	**0.00**	0.045
SG2B	11	24	1	5	4.00	31.25	0.379
WG7B	13	24	1	3	4.00	18.75	0.376
LG8B	12	25	0	4	**0.00**	25.00	0.077

Table 82: Toxic Language Hate Speech - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	56.10	52.00	45.45	62.50	52.63	14.18
GG2B	**90.24**	84.00	80.00	**100.00**	**88.89**	**81.98**
SG2B	85.37	96.00	91.67	68.75	78.57	69.42
WG7B	**90.24**	96.00	92.86	81.25	86.67	79.47
LG8B	**90.24**	**100.00**	**100.00**	75.00	85.71	80.41

Table 83: Social Stereotypes and Unfair Discrimination - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	13	15	32	16	68.09	55.17	0.049
GG2B	20	42	5	9	10.64	31.03	0.044
SG2B	14	42	5	15	10.64	51.72	0.378
WG7B	21	43	4	8	8.51	**27.59**	0.356
LG8B	16	46	1	13	**2.13**	44.83	0.068

Table 84: Social Stereotypes and Unfair Discrimination - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	36.84	31.91	28.89	44.83	35.14	-22.99
GG2B	81.58	89.36	80.00	68.97	74.07	60.31
SG2B	73.68	89.36	73.68	48.28	58.33	42.22
WG7B	**84.21**	91.49	84.00	**72.41**	**77.78**	**66.07**
LG8B	81.58	**97.87**	**94.12**	55.17	69.57	61.84

Table 85: Mental Health Over-Reliance Crisis - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	10	17	14	6	45.16	37.50	0.048
GG2B	15	27	4	1	12.90	**6.25**	0.044
SG2B	8	31	0	8	**0.00**	50.00	0.387
WG7B	11	31	0	5	**0.00**	31.25	0.359
LG8B	11	28	3	5	9.68	31.25	0.076

Table 86: Mental Health Over-Reliance Crisis - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	57.45	54.84	41.67	62.50	50.00	16.44
GG2B	**89.36**	87.10	78.95	**93.75**	**85.71**	**78.06**
SG2B	82.98	**100.00**	**100.00**	50.00	66.67	63.04
WG7B	**89.36**	**100.00**	**100.00**	68.75	81.48	76.94
LG8B	82.98	90.32	78.57	68.75	73.33	61.21

Table 87: Fraud Assisting Illegal Activities - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	12	17	19	12	52.78	50.00	0.048
GG2B	19	32	4	5	11.11	**20.83**	0.044
SG2B	10	34	2	14	5.56	58.33	0.374
WG7B	18	35	1	6	**2.78**	25.00	0.359
LG8B	19	32	4	5	11.11	**20.83**	0.076

Table 88: Fraud Assisting Illegal Activities - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	48.33	47.22	38.71	50.00	43.64	-2.72
GG2B	85.00	88.89	82.61	**79.17**	80.85	68.57
SG2B	73.33	94.44	83.33	41.67	55.56	44.23
WG7B	**88.33**	**97.22**	**94.74**	75.00	**83.72**	**76.06**
LG8B	85.00	88.89	82.61	**79.17**	80.85	68.57

Table 89: Defamation Encouraging Unethical or Unsafe Actions - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	9	13	16	8	55.17	47.06	0.048
GG2B	15	25	4	2	13.79	**11.76**	0.044
SG2B	5	26	3	12	10.34	70.59	0.378
WG7B	14	27	2	3	**6.90**	17.65	0.358
LG8B	14	26	3	3	10.34	17.65	0.076

Table 90: Defamation Encouraging Unethical or Unsafe Actions - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	47.83	44.83	36.00	52.94	42.86	-2.16
GG2B	86.96	86.21	78.95	**88.24**	83.33	72.98
SG2B	67.39	89.66	62.50	29.41	40.00	24.28
WG7B	**89.13**	**93.10**	**87.50**	82.35	**84.85**	**76.47**
LG8B	86.96	89.66	82.35	82.35	82.35	72.01

Table 91: Benign - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	10	413	535	13	56.43	56.52	0.049
GG2B	11	910	38	12	4.01	**52.17**	0.045
SG2B	3	942	6	20	**0.63**	86.96	0.384
WG7B	6	937	11	17	1.16	73.91	0.361
LG8B	2	934	14	21	1.48	91.30	0.066

Table 92: Benign - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	43.56	43.57	1.83	43.48	3.52	-3.97
GG2B	94.85	95.99	22.45	**47.83**	**30.56**	**30.44**
SG2B	**97.32**	**99.37**	33.33	13.04	18.75	19.69
WG7B	97.12	98.84	**35.29**	26.09	30.00	28.90
LG8B	96.40	98.52	12.50	8.70	10.26	8.62

Table 93: Others - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	22	26	33	17	55.93	43.59	0.046
GG2B	29	57	2	10	3.39	**25.64**	0.043
SG2B	18	56	3	21	5.08	53.85	0.382
WG7B	26	58	1	13	1.69	33.33	0.35
LG8B	26	59	0	13	**0.00**	33.33	0.066

Table 94: Others - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	48.98	44.07	40.00	56.41	46.81	0.47
GG2B	**87.76**	96.61	93.55	**74.36**	**82.86**	**74.70**
SG2B	75.51	94.92	85.71	46.15	60.00	48.99
WG7B	85.71	98.31	96.30	66.67	78.79	71.18
LG8B	86.73	**100.00**	**100.00**	66.67	80.00	73.91

Table 95: Violence and Physical Harm - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	5	11	11	13	50.00	72.22	0.045
GG2B	15	19	3	3	13.64	**16.67**	0.045
SG2B	10	22	0	8	**0.00**	44.44	0.377
WG7B	15	22	0	3	**0.00**	**16.67**	0.36
LG8B	13	22	0	5	**0.00**	27.78	0.073

Table 96: Violence and Physical Harm - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	40.00	50.00	31.25	27.78	29.41	-22.57
GG2B	85.00	86.36	83.33	**83.33**	83.33	69.70
SG2B	80.00	**100.00**	**100.00**	55.56	71.43	63.83
WG7B	**92.50**	**100.00**	**100.00**	**83.33**	**90.91**	**85.63**
LG8B	87.50	**100.00**	**100.00**	72.22	83.87	76.71

Table 97: Copyright Violations - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	2	11	13	5	54.17	71.43	0.046
GG2B	5	23	1	2	4.17	**28.57**	0.044
SG2B	1	24	0	6	**0.00**	85.71	0.384
WG7B	2	21	3	5	12.50	71.43	0.358
LG8B	5	20	4	2	16.67	**28.57**	0.076

Table 98: Copyright Violations - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	41.94	45.83	13.33	28.57	18.18	-21.41
GG2B	**90.32**	95.83	83.33	**71.43**	**76.92**	**71.18**
SG2B	80.65	**100.00**	**100.00**	14.29	25.00	33.81
WG7B	74.19	87.50	40.00	28.57	33.33	18.27
LG8B	80.65	83.33	55.56	**71.43**	62.50	50.44

Table 99: Sensitive Information Organization Government - Table 1

Model	TP	TN	FP	FN	FPR(%)	FNR(%)	AvgTime (s)
LG1B	10	17	15	8	46.88	**44.44**	0.047
GG2B	9	31	1	9	3.12	50.00	0.044
SG2B	0	32	0	18	**0.00**	100.00	0.384
WG7B	6	32	0	12	**0.00**	66.67	0.374
LG8B	8	32	0	10	**0.00**	55.56	0.068

Table 100: Sensitive Information Organization Government - Table 2

Model	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	54.00	53.12	40.00	**55.56**	46.51	8.33
GG2B	**80.00**	96.88	90.00	50.00	**64.29**	56.25
SG2B	64.00	**100.00**	nan	0.00	nan	nan
WG7B	76.00	**100.00**	**100.00**	33.33	50.00	49.24
LG8B	**80.00**	**100.00**	**100.00**	44.44	61.54	**58.19**

3 Single-model VS Combined-models Tables

3.1 XSTest

Table 101: XSTest - Table 1

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B	-	87	145	105	113	42.00	56.50
GG2B	-	198	169	81	2	32.40	1.00
SG2B	-	165	196	54	35	21.60	17.50
WG7B	-	183	247	3	17	1.20	8.50
LG8B	-	164	243	7	36	2.80	18.00
LG1B/GG2B	OR	198	97	153	2	61.20	1.00
LG1B/GG2B	AND	87	217	33	113	13.20	56.50
LG1B/GG2B	AVG	198	97	153	2	61.20	1.00
LG1B/SG2B	OR	181	110	140	19	56.00	9.50
LG1B/SG2B	AND	71	231	19	129	7.60	64.50
LG1B/SG2B	AVG	181	110	140	19	56.00	9.50
LG1B/WG7B	OR	187	143	107	13	42.80	6.50
LG1B/WG7B	AND	83	249	1	117	0.40	58.50
LG1B/WG7B	AVG	187	143	107	13	42.80	6.50
LG1B/LG8B	OR	177	142	108	23	43.20	11.50
LG1B/LG8B	AND	74	246	4	126	1.60	63.00
LG1B/LG8B	AVG	177	142	108	23	43.20	11.50
GG2B/SG2B	OR	199	149	101	1	40.40	0.50
GG2B/SG2B	AND	164	216	34	36	13.60	18.00
GG2B/SG2B	AVG	199	149	101	1	40.40	0.50
GG2B/WG7B	OR	199	168	82	1	32.80	0.50
GG2B/WG7B	AND	182	248	2	18	0.80	9.00
GG2B/WG7B	AVG	199	168	82	1	32.80	0.50
GG2B/LG8B	OR	198	169	81	2	32.40	1.00
GG2B/LG8B	AND	164	243	7	36	2.80	18.00
GG2B/LG8B	AVG	198	169	81	2	32.40	1.00
SG2B/WG7B	OR	194	195	55	6	22.00	3.00
SG2B/WG7B	AND	154	248	2	46	0.80	23.00
SG2B/WG7B	AVG	194	195	55	6	22.00	3.00
SG2B/LG8B	OR	185	191	59	15	23.60	7.50
SG2B/LG8B	AND	144	248	2	56	0.80	28.00
SG2B/LG8B	AVG	185	191	59	15	23.60	7.50
WG7B/LG8B	OR	188	240	10	12	4.00	6.00
WG7B/LG8B	AND	159	250	0	41	**0.00**	20.50
WG7B/LG8B	AVG	188	240	10	12	4.00	6.00
LG1B/GG2B/SG2B	OR	199	85	165	1	66.00	0.50
LG1B/GG2B/SG2B	AND	71	239	11	129	4.40	64.50
LG1B/GG2B/SG2B	AVG	180	186	64	20	25.60	10.00
LG1B/GG2B/WG7B	OR	199	97	153	1	61.20	0.50
LG1B/GG2B/WG7B	AND	83	250	0	117	**0.00**	58.50
LG1B/GG2B/WG7B	AVG	186	214	36	14	14.40	7.00
LG1B/GG2B/LG8B	OR	198	97	153	2	61.20	1.00
LG1B/GG2B/LG8B	AND	74	246	4	126	1.60	63.00

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Table 101 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B/GG2B/LG8B	AVG	177	214	36	23	14.40	11.50
LG1B/SG2B/WG7B	OR	195	110	140	5	56.00	2.50
LG1B/SG2B/WG7B	AND	68	250	0	132	**0.00**	66.00
LG1B/SG2B/WG7B	AVG	172	228	22	28	8.80	14.00
LG1B/SG2B/LG8B	OR	189	108	142	11	56.80	5.50
LG1B/SG2B/LG8B	AND	62	249	1	138	0.40	69.00
LG1B/SG2B/LG8B	AVG	165	227	23	35	9.20	17.50
LG1B/WG7B/LG8B	OR	191	140	110	9	44.00	4.50
LG1B/WG7B/LG8B	AND	73	250	0	127	**0.00**	63.50
LG1B/WG7B/LG8B	AVG	170	245	5	30	2.00	15.00
GG2B/SG2B/WG7B	OR	200	148	102	0	40.80	**0.00**
GG2B/SG2B/WG7B	AND	154	248	2	46	0.80	23.00
GG2B/SG2B/WG7B	AVG	192	216	34	8	13.60	4.00
GG2B/SG2B/LG8B	OR	199	149	101	1	40.40	0.50
GG2B/SG2B/LG8B	AND	144	248	2	56	0.80	28.00
GG2B/SG2B/LG8B	AVG	184	211	39	16	15.60	8.00
GG2B/WG7B/LG8B	OR	199	168	82	1	32.80	0.50
GG2B/WG7B/LG8B	AND	159	250	0	41	**0.00**	20.50
GG2B/WG7B/LG8B	AVG	187	241	9	13	3.60	6.50
SG2B/WG7B/LG8B	OR	196	190	60	4	24.00	2.00
SG2B/WG7B/LG8B	AND	141	250	0	59	**0.00**	29.50
SG2B/WG7B/LG8B	AVG	175	246	4	25	1.60	12.50
LG1B/GG2B/SG2B/WG7B	OR	200	85	165	0	66.00	**0.00**
LG1B/GG2B/SG2B/WG7B	AND	68	250	0	132	**0.00**	66.00
LG1B/GG2B/SG2B/WG7B	AVG	193	185	65	7	26.00	3.50
LG1B/GG2B/SG2B/LG8B	OR	199	85	165	1	66.00	0.50
LG1B/GG2B/SG2B/LG8B	AND	62	249	1	138	0.40	69.00
LG1B/GG2B/SG2B/LG8B	AVG	188	184	66	12	26.40	6.00
LG1B/GG2B/WG7B/LG8B	OR	199	97	153	1	61.20	0.50
LG1B/GG2B/WG7B/LG8B	AND	73	250	0	127	**0.00**	63.50
LG1B/GG2B/WG7B/LG8B	AVG	190	211	39	10	15.60	5.00
LG1B/SG2B/WG7B/LG8B	OR	196	108	142	4	56.80	2.00
LG1B/SG2B/WG7B/LG8B	AND	62	250	0	138	**0.00**	69.00
LG1B/SG2B/WG7B/LG8B	AVG	183	224	26	17	10.40	8.50
GG2B/SG2B/WG7B/LG8B	OR	200	148	102	0	40.80	**0.00**
GG2B/SG2B/WG7B/LG8B	AND	141	250	0	59	**0.00**	29.50
GG2B/SG2B/WG7B/LG8B	AVG	194	211	39	6	15.60	3.00
LG1B/GG2B/SG2B/WG7B/LG8B	OR	200	85	165	0	66.00	**0.00**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	62	250	0	138	**0.00**	69.00
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	183	233	17	17	6.80	8.50

Table 102: XSTest - Table 2

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	-	51.56	58.00	45.31	43.50	44.39	1.51
GG2B	-	81.56	67.60	70.97	99.00	82.67	68.18
SG2B	-	80.22	78.40	75.34	82.50	78.76	60.54
WG7B	-	**95.56**	98.80	98.39	91.50	**94.82**	91.12
LG8B	-	90.44	97.20	95.91	82.00	88.41	81.08
LG1B/GG2B	OR	65.56	38.80	56.41	99.00	71.87	45.34
LG1B/GG2B	AND	67.56	86.80	72.50	43.50	54.38	34.05
LG1B/GG2B	AVG	65.56	38.80	56.41	99.00	71.87	45.34
LG1B/SG2B	OR	64.67	44.00	56.39	90.50	69.48	37.91
LG1B/SG2B	AND	67.11	92.40	78.89	35.50	48.97	34.66
LG1B/SG2B	AVG	64.67	44.00	56.39	90.50	69.48	37.91
LG1B/WG7B	OR	73.33	57.20	63.61	93.50	75.71	52.94
LG1B/WG7B	AND	73.78	99.60	98.81	41.50	58.45	52.41
LG1B/WG7B	AVG	73.33	57.20	63.61	93.50	75.71	52.94
LG1B/LG8B	OR	70.89	56.80	62.11	88.50	72.99	46.71
LG1B/LG8B	AND	71.11	98.40	94.87	37.00	53.24	46.47
LG1B/LG8B	AVG	70.89	56.80	62.11	88.50	72.99	46.71
GG2B/SG2B	OR	77.33	59.60	66.33	99.50	79.60	62.30
GG2B/SG2B	AND	84.44	86.40	82.83	82.00	82.41	68.47
GG2B/SG2B	AVG	77.33	59.60	66.33	99.50	79.60	62.30
GG2B/WG7B	OR	81.56	67.20	70.82	99.50	82.74	68.44
GG2B/WG7B	AND	**95.56**	99.20	98.91	91.00	94.79	**91.17**
GG2B/WG7B	AVG	81.56	67.20	70.82	99.50	82.74	68.44
GG2B/LG8B	OR	81.56	67.60	70.97	99.00	82.67	68.18
GG2B/LG8B	AND	90.44	97.20	95.91	82.00	88.41	81.08
GG2B/LG8B	AVG	81.56	67.60	70.97	99.00	82.67	68.18
SG2B/WG7B	OR	86.44	78.00	77.91	97.00	86.41	74.96
SG2B/WG7B	AND	89.33	99.20	98.72	77.00	86.52	79.56
SG2B/WG7B	AVG	86.44	78.00	77.91	97.00	86.41	74.96
SG2B/LG8B	OR	83.56	76.40	75.82	92.50	83.33	68.72
SG2B/LG8B	AND	87.11	99.20	98.63	72.00	83.24	75.57
SG2B/LG8B	AVG	83.56	76.40	75.82	92.50	83.33	68.72
WG7B/LG8B	OR	95.11	96.00	94.95	94.00	94.47	90.09
WG7B/LG8B	AND	90.89	**100.00**	**100.00**	79.50	88.58	82.64
WG7B/LG8B	AVG	95.11	96.00	94.95	94.00	94.47	90.09
LG1B/GG2B/SG2B	OR	63.11	34.00	54.67	99.50	70.57	42.34
LG1B/GG2B/SG2B	AND	68.89	95.60	86.59	35.50	50.35	40.03
LG1B/GG2B/SG2B	AVG	81.33	74.40	73.77	90.00	81.08	64.23
LG1B/GG2B/WG7B	OR	65.78	38.80	56.53	99.50	72.10	46.11
LG1B/GG2B/WG7B	AND	74.00	**100.00**	**100.00**	41.50	58.66	53.17
LG1B/GG2B/WG7B	AVG	88.89	85.60	83.78	93.00	88.15	78.12
LG1B/GG2B/LG8B	OR	65.56	38.80	56.41	99.00	71.87	45.34
LG1B/GG2B/LG8B	AND	71.11	98.40	94.87	37.00	53.24	46.47
LG1B/GG2B/LG8B	AVG	86.89	85.60	83.10	88.50	85.71	73.75
LG1B/SG2B/WG7B	OR	67.78	44.00	58.21	97.50	72.90	47.28

Continued on next page

Table 102 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B/SG2B/WG7B	AND	70.67	**100.00**	**100.00**	34.00	50.75	47.17
LG1B/SG2B/WG7B	AVG	88.89	91.20	88.66	86.00	87.31	77.46
LG1B/SG2B/LG8B	OR	66.00	43.20	57.10	94.50	71.19	42.48
LG1B/SG2B/LG8B	AND	69.11	99.60	98.41	31.00	47.15	43.82
LG1B/SG2B/LG8B	AVG	87.11	90.80	87.77	82.50	85.05	73.85
LG1B/WG7B/LG8B	OR	73.56	56.00	63.46	95.50	76.25	54.38
LG1B/WG7B/LG8B	AND	71.78	**100.00**	**100.00**	36.50	53.48	49.20
LG1B/WG7B/LG8B	AVG	92.22	98.00	97.14	85.00	90.67	84.60
GG2B/SG2B/WG7B	OR	77.33	59.20	66.23	**100.00**	79.68	62.61
GG2B/SG2B/WG7B	AND	89.33	99.20	98.72	77.00	86.52	79.56
GG2B/SG2B/WG7B	AVG	90.67	86.40	84.96	96.00	90.14	81.89
GG2B/SG2B/LG8B	OR	77.33	59.60	66.33	99.50	79.60	62.30
GG2B/SG2B/LG8B	AND	87.11	99.20	98.63	72.00	83.24	75.57
GG2B/SG2B/LG8B	AVG	87.78	84.40	82.51	92.00	87.00	75.93
GG2B/WG7B/LG8B	OR	81.56	67.20	70.82	99.50	82.74	68.44
GG2B/WG7B/LG8B	AND	90.89	**100.00**	**100.00**	79.50	88.58	82.64
GG2B/WG7B/LG8B	AVG	95.11	96.40	95.41	93.50	94.44	90.09
SG2B/WG7B/LG8B	OR	85.78	76.00	76.56	98.00	85.96	74.25
SG2B/WG7B/LG8B	AND	86.89	**100.00**	**100.00**	70.50	82.70	75.52
SG2B/WG7B/LG8B	AVG	93.56	98.40	97.77	87.50	92.35	87.21
LG1B/GG2B/SG2B/WG7B	OR	63.33	34.00	54.79	**100.00**	70.80	43.16
LG1B/GG2B/SG2B/WG7B	AND	70.67	**100.00**	**100.00**	34.00	50.75	47.17
LG1B/GG2B/SG2B/WG7B	AVG	84.00	74.00	74.81	96.50	84.28	70.83
LG1B/GG2B/SG2B/LG8B	OR	63.11	34.00	54.67	99.50	70.57	42.34
LG1B/GG2B/SG2B/LG8B	AND	69.11	99.60	98.41	31.00	47.15	43.82
LG1B/GG2B/SG2B/LG8B	AVG	82.67	73.60	74.02	94.00	82.82	67.75
LG1B/GG2B/WG7B/LG8B	OR	65.78	38.80	56.53	99.50	72.10	46.11
LG1B/GG2B/WG7B/LG8B	AND	71.78	**100.00**	**100.00**	36.50	53.48	49.20
LG1B/GG2B/WG7B/LG8B	AVG	89.11	84.40	82.97	95.00	88.58	78.92
LG1B/SG2B/WG7B/LG8B	OR	67.56	43.20	57.99	98.00	72.86	47.35
LG1B/SG2B/WG7B/LG8B	AND	69.33	**100.00**	**100.00**	31.00	47.33	44.69
LG1B/SG2B/WG7B/LG8B	AVG	90.44	89.60	87.56	91.50	89.49	80.80
GG2B/SG2B/WG7B/LG8B	OR	77.33	59.20	66.23	**100.00**	79.68	62.61
GG2B/SG2B/WG7B/LG8B	AND	86.89	**100.00**	**100.00**	70.50	82.70	75.52
GG2B/SG2B/WG7B/LG8B	AVG	90.00	84.40	83.26	97.00	89.61	80.95
LG1B/GG2B/SG2B/WG7B/LG8B	OR	63.33	34.00	54.79	**100.00**	70.80	43.16
LG1B/GG2B/SG2B/WG7B/LG8B	AND	69.33	**100.00**	**100.00**	31.00	47.33	44.69
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	92.44	93.20	91.50	91.50	91.50	84.70

3.2 MedSafetyBench

Table 103: MedSafetyBench - Table 1

Model	Logic	TP	FN	FNR(%)
LG1B	-	202	248	55.11
GG2B	-	395	55	12.22
SG2B	-	114	336	74.67
WG7B	-	396	54	12.00
LG8B	-	231	219	48.67
LG1B/GG2B	OR	420	30	6.67
LG1B/GG2B	AND	177	273	60.67
LG1B/GG2B	AVG	420	30	6.67
LG1B/SG2B	OR	266	184	40.89
LG1B/SG2B	AND	50	400	88.89
LG1B/SG2B	AVG	266	184	40.89
LG1B/WG7B	OR	423	27	6.00
LG1B/WG7B	AND	175	275	61.11
LG1B/WG7B	AVG	423	27	6.00
LG1B/LG8B	OR	332	118	26.22
LG1B/LG8B	AND	101	349	77.56
LG1B/LG8B	AVG	332	118	26.22
GG2B/SG2B	OR	396	54	12.00
GG2B/SG2B	AND	113	337	74.89
GG2B/SG2B	AVG	396	54	12.00
GG2B/WG7B	OR	412	38	8.44
GG2B/WG7B	AND	379	71	15.78
GG2B/WG7B	AVG	412	38	8.44
GG2B/LG8B	OR	396	54	12.00
GG2B/LG8B	AND	230	220	48.89
GG2B/LG8B	AVG	396	54	12.00
SG2B/WG7B	OR	398	52	11.56
SG2B/WG7B	AND	112	338	75.11
SG2B/WG7B	AVG	398	52	11.56
SG2B/LG8B	OR	262	188	41.78
SG2B/LG8B	AND	83	367	81.56
SG2B/LG8B	AVG	262	188	41.78
WG7B/LG8B	OR	397	53	11.78
WG7B/LG8B	AND	230	220	48.89
WG7B/LG8B	AVG	397	53	11.78
LG1B/GG2B/SG2B	OR	421	29	6.44
LG1B/GG2B/SG2B	AND	50	400	88.89
LG1B/GG2B/SG2B	AVG	240	210	46.67
LG1B/GG2B/WG7B	OR	430	20	**4.44**
LG1B/GG2B/WG7B	AND	168	282	62.67
LG1B/GG2B/WG7B	AVG	395	55	12.22
LG1B/GG2B/LG8B	OR	421	29	6.44
LG1B/GG2B/LG8B	AND	101	349	77.56
LG1B/GG2B/LG8B	AVG	306	144	32.00
LG1B/SG2B/WG7B	OR	425	25	5.56

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Table 103 – continued from previous page

Model	Logic	TP	FN	FNR(%)
LG1B/SG2B/WG7B	AND	50	400	88.89
LG1B/SG2B/WG7B	AVG	237	213	47.33
LG1B/SG2B/LG8B	OR	350	100	22.22
LG1B/SG2B/LG8B	AND	37	413	91.78
LG1B/SG2B/LG8B	AVG	160	290	64.44
LG1B/WG7B/LG8B	OR	424	26	5.78
LG1B/WG7B/LG8B	AND	101	349	77.56
LG1B/WG7B/LG8B	AVG	304	146	32.44
GG2B/SG2B/WG7B	OR	412	38	8.44
GG2B/SG2B/WG7B	AND	111	339	75.33
GG2B/SG2B/WG7B	AVG	382	68	15.11
GG2B/SG2B/LG8B	OR	397	53	11.78
GG2B/SG2B/LG8B	AND	83	367	81.56
GG2B/SG2B/LG8B	AVG	260	190	42.22
GG2B/WG7B/LG8B	OR	412	38	8.44
GG2B/WG7B/LG8B	AND	229	221	49.11
GG2B/WG7B/LG8B	AVG	381	69	15.33
SG2B/WG7B/LG8B	OR	398	52	11.56
SG2B/WG7B/LG8B	AND	82	368	81.78
SG2B/WG7B/LG8B	AVG	261	189	42.00
LG1B/GG2B/SG2B/WG7B	OR	430	20	**4.44**
LG1B/GG2B/SG2B/WG7B	AND	50	400	88.89
LG1B/GG2B/SG2B/WG7B	AVG	398	52	11.56
LG1B/GG2B/SG2B/LG8B	OR	422	28	6.22
LG1B/GG2B/SG2B/LG8B	AND	37	413	91.78
LG1B/GG2B/SG2B/LG8B	AVG	323	127	28.22
LG1B/GG2B/WG7B/LG8B	OR	430	20	**4.44**
LG1B/GG2B/WG7B/LG8B	AND	101	349	77.56
LG1B/GG2B/WG7B/LG8B	AVG	397	53	11.78
LG1B/SG2B/WG7B/LG8B	OR	425	25	5.56
LG1B/SG2B/WG7B/LG8B	AND	37	413	91.78
LG1B/SG2B/WG7B/LG8B	AVG	322	128	28.44
GG2B/SG2B/WG7B/LG8B	OR	412	38	8.44
GG2B/SG2B/WG7B/LG8B	AND	82	368	81.78
GG2B/SG2B/WG7B/LG8B	AVG	383	67	14.89
LG1B/GG2B/SG2B/WG7B/LG8B	OR	430	20	**4.44**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	37	413	91.78
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	313	137	30.44

Table 104: MedSafetyBench - Table 2

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	-	44.89	44.89	61.96
GG2B	-	87.78	87.78	93.49

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Table 104 – continued from previous page

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
SG2B	-	25.33	25.33	40.43
WG7B	-	88.00	88.00	93.62
LG8B	-	51.33	51.33	67.84
LG1B/GG2B	OR	93.33	93.33	96.55
LG1B/GG2B	AND	39.33	39.33	56.46
LG1B/GG2B	AVG	93.33	93.33	96.55
LG1B/SG2B	OR	59.11	59.11	74.30
LG1B/SG2B	AND	11.11	11.11	20.00
LG1B/SG2B	AVG	59.11	59.11	74.30
LG1B/WG7B	OR	94.00	94.00	96.91
LG1B/WG7B	AND	38.89	38.89	56.00
LG1B/WG7B	AVG	94.00	94.00	96.91
LG1B/LG8B	OR	73.78	73.78	84.91
LG1B/LG8B	AND	22.44	22.44	36.66
LG1B/LG8B	AVG	73.78	73.78	84.91
GG2B/SG2B	OR	88.00	88.00	93.62
GG2B/SG2B	AND	25.11	25.11	40.14
GG2B/SG2B	AVG	88.00	88.00	93.62
GG2B/WG7B	OR	91.56	91.56	95.59
GG2B/WG7B	AND	84.22	84.22	91.44
GG2B/WG7B	AVG	91.56	91.56	95.59
GG2B/LG8B	OR	88.00	88.00	93.62
GG2B/LG8B	AND	51.11	51.11	67.65
GG2B/LG8B	AVG	88.00	88.00	93.62
SG2B/WG7B	OR	88.44	88.44	93.87
SG2B/WG7B	AND	24.89	24.89	39.86
SG2B/WG7B	AVG	88.44	88.44	93.87
SG2B/LG8B	OR	58.22	58.22	73.60
SG2B/LG8B	AND	18.44	18.44	31.14
SG2B/LG8B	AVG	58.22	58.22	73.60
WG7B/LG8B	OR	88.22	88.22	93.74
WG7B/LG8B	AND	51.11	51.11	67.65
WG7B/LG8B	AVG	88.22	88.22	93.74
LG1B/GG2B/SG2B	OR	93.56	93.56	96.67
LG1B/GG2B/SG2B	AND	11.11	11.11	20.00
LG1B/GG2B/SG2B	AVG	53.33	53.33	69.57
LG1B/GG2B/WG7B	OR	**95.56**	**95.56**	**97.73**
LG1B/GG2B/WG7B	AND	37.33	37.33	54.37
LG1B/GG2B/WG7B	AVG	87.78	87.78	93.49
LG1B/GG2B/LG8B	OR	93.56	93.56	96.67
LG1B/GG2B/LG8B	AND	22.44	22.44	36.66
LG1B/GG2B/LG8B	AVG	68.00	68.00	80.95
LG1B/SG2B/WG7B	OR	94.44	94.44	97.14
LG1B/SG2B/WG7B	AND	11.11	11.11	20.00
LG1B/SG2B/WG7B	AVG	52.67	52.67	69.00
LG1B/SG2B/LG8B	OR	77.78	77.78	87.50

Continued on next page

Table 104 – continued from previous page

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
LG1B/SG2B/LG8B	AND	8.22	8.22	15.20
LG1B/SG2B/LG8B	AVG	35.56	35.56	52.46
LG1B/WG7B/LG8B	OR	94.22	94.22	97.03
LG1B/WG7B/LG8B	AND	22.44	22.44	36.66
LG1B/WG7B/LG8B	AVG	67.56	67.56	80.64
GG2B/SG2B/WG7B	OR	91.56	91.56	95.59
GG2B/SG2B/WG7B	AND	24.67	24.67	39.57
GG2B/SG2B/WG7B	AVG	84.89	84.89	91.83
GG2B/SG2B/LG8B	OR	88.22	88.22	93.74
GG2B/SG2B/LG8B	AND	18.44	18.44	31.14
GG2B/SG2B/LG8B	AVG	57.78	57.78	73.24
GG2B/WG7B/LG8B	OR	91.56	91.56	95.59
GG2B/WG7B/LG8B	AND	50.89	50.89	67.45
GG2B/WG7B/LG8B	AVG	84.67	84.67	91.70
SG2B/WG7B/LG8B	OR	88.44	88.44	93.87
SG2B/WG7B/LG8B	AND	18.22	18.22	30.83
SG2B/WG7B/LG8B	AVG	58.00	58.00	73.42
LG1B/GG2B/SG2B/WG7B	OR	**95.56**	**95.56**	**97.73**
LG1B/GG2B/SG2B/WG7B	AND	11.11	11.11	20.00
LG1B/GG2B/SG2B/WG7B	AVG	88.44	88.44	93.87
LG1B/GG2B/SG2B/LG8B	OR	93.78	93.78	96.79
LG1B/GG2B/SG2B/LG8B	AND	8.22	8.22	15.20
LG1B/GG2B/SG2B/LG8B	AVG	71.78	71.78	83.57
LG1B/GG2B/WG7B/LG8B	OR	**95.56**	**95.56**	**97.73**
LG1B/GG2B/WG7B/LG8B	AND	22.44	22.44	36.66
LG1B/GG2B/WG7B/LG8B	AVG	88.22	88.22	93.74
LG1B/SG2B/WG7B/LG8B	OR	94.44	94.44	97.14
LG1B/SG2B/WG7B/LG8B	AND	8.22	8.22	15.20
LG1B/SG2B/WG7B/LG8B	AVG	71.56	71.56	83.42
GG2B/SG2B/WG7B/LG8B	OR	91.56	91.56	95.59
GG2B/SG2B/WG7B/LG8B	AND	18.22	18.22	30.83
GG2B/SG2B/WG7B/LG8B	AVG	85.11	85.11	91.96
LG1B/GG2B/SG2B/WG7B/LG8B	OR	**95.56**	**95.56**	**97.73**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	8.22	8.22	15.20
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	69.56	69.56	82.04

3.3 PatientSafetyBench

Table 105: PatientSafetyBench - Table 1

Model	Logic	TP	FN	FNR(%)
LG1B	-	204	262	56.22
GG2B	-	234	232	49.79
SG2B	-	125	341	73.18

Continued on next page

Table 105 – continued from previous page

Model	Logic	TP	FN	FNR(%)
WG7B	-	208	258	55.36
LG8B	-	194	272	58.37
LG1B/GG2B	OR	334	132	28.33
LG1B/GG2B	AND	104	362	77.68
LG1B/GG2B	AVG	334	132	28.33
LG1B/SG2B	OR	269	197	42.27
LG1B/SG2B	AND	60	406	87.12
LG1B/SG2B	AVG	269	197	42.27
LG1B/WG7B	OR	319	147	31.55
LG1B/WG7B	AND	93	373	80.04
LG1B/WG7B	AVG	319	147	31.55
LG1B/LG8B	OR	319	147	31.55
LG1B/LG8B	AND	79	387	83.05
LG1B/LG8B	AVG	319	147	31.55
GG2B/SG2B	OR	240	226	48.50
GG2B/SG2B	AND	119	347	74.46
GG2B/SG2B	AVG	240	226	48.50
GG2B/WG7B	OR	270	196	42.06
GG2B/WG7B	AND	172	294	63.09
GG2B/WG7B	AVG	270	196	42.06
GG2B/LG8B	OR	301	165	35.41
GG2B/LG8B	AND	127	339	72.75
GG2B/LG8B	AVG	301	165	35.41
SG2B/WG7B	OR	232	234	50.21
SG2B/WG7B	AND	101	365	78.33
SG2B/WG7B	AVG	232	234	50.21
SG2B/LG8B	OR	272	194	41.63
SG2B/LG8B	AND	47	419	89.91
SG2B/LG8B	AVG	272	194	41.63
WG7B/LG8B	OR	288	178	38.20
WG7B/LG8B	AND	114	352	75.54
WG7B/LG8B	AVG	288	178	38.20
LG1B/GG2B/SG2B	OR	337	129	27.68
LG1B/GG2B/SG2B	AND	57	409	87.77
LG1B/GG2B/SG2B	AVG	169	297	63.73
LG1B/GG2B/WG7B	OR	355	111	23.82
LG1B/GG2B/WG7B	AND	78	388	83.26
LG1B/GG2B/WG7B	AVG	213	253	54.29
LG1B/GG2B/LG8B	OR	374	92	19.74
LG1B/GG2B/LG8B	AND	52	414	88.84
LG1B/GG2B/LG8B	AVG	206	260	55.79
LG1B/SG2B/WG7B	OR	331	135	28.97
LG1B/SG2B/WG7B	AND	48	418	89.70
LG1B/SG2B/WG7B	AVG	158	308	66.09
LG1B/SG2B/LG8B	OR	357	109	23.39
LG1B/SG2B/LG8B	AND	20	446	95.71

Continued on next page

Table 105 – continued from previous page

Model	Logic	TP	FN	FNR(%)
LG1B/SG2B/LG8B	AVG	146	320	68.67
LG1B/WG7B/LG8B	OR	369	97	20.82
LG1B/WG7B/LG8B	AND	49	417	89.48
LG1B/WG7B/LG8B	AVG	188	278	59.66
GG2B/SG2B/WG7B	OR	273	193	41.42
GG2B/SG2B/WG7B	AND	98	368	78.97
GG2B/SG2B/WG7B	AVG	196	270	57.94
GG2B/SG2B/LG8B	OR	306	160	34.33
GG2B/SG2B/LG8B	AND	46	420	90.13
GG2B/SG2B/LG8B	AVG	201	265	56.87
GG2B/WG7B/LG8B	OR	323	143	30.69
GG2B/WG7B/LG8B	AND	100	366	78.54
GG2B/WG7B/LG8B	AVG	213	253	54.29
SG2B/WG7B/LG8B	OR	309	157	33.69
SG2B/WG7B/LG8B	AND	44	422	90.56
SG2B/WG7B/LG8B	AVG	174	292	62.66
LG1B/GG2B/SG2B/WG7B	OR	356	110	23.61
LG1B/GG2B/SG2B/WG7B	AND	47	419	89.91
LG1B/GG2B/SG2B/WG7B	AVG	228	238	51.07
LG1B/GG2B/SG2B/LG8B	OR	377	89	19.10
LG1B/GG2B/SG2B/LG8B	AND	19	447	95.92
LG1B/GG2B/SG2B/LG8B	AVG	243	223	47.85
LG1B/GG2B/WG7B/LG8B	OR	386	80	17.17
LG1B/GG2B/WG7B/LG8B	AND	44	422	90.56
LG1B/GG2B/WG7B/LG8B	AVG	263	203	43.56
LG1B/SG2B/WG7B/LG8B	OR	379	87	18.67
LG1B/SG2B/WG7B/LG8B	AND	19	447	95.92
LG1B/SG2B/WG7B/LG8B	AVG	229	237	50.86
GG2B/SG2B/WG7B/LG8B	OR	325	141	30.26
GG2B/SG2B/WG7B/LG8B	AND	44	422	90.56
GG2B/SG2B/WG7B/LG8B	AVG	236	230	49.36
LG1B/GG2B/SG2B/WG7B/LG8B	OR	387	79	**16.95**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	19	447	95.92
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	187	279	59.87

Table 106: PatientSafetyBench - Table 2

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
LG1B	-	43.78	43.78	60.90
GG2B	-	50.21	50.21	66.86
SG2B	-	26.82	26.82	42.30
WG7B	-	44.64	44.64	61.72
LG8B	-	41.63	41.63	58.79
LG1B/GG2B	OR	71.67	71.67	83.50

Continued on next page

Table 106 – continued from previous page

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
LG1B/GG2B	AND	22.32	22.32	36.49
LG1B/GG2B	AVG	71.67	71.67	83.50
LG1B/SG2B	OR	57.73	57.73	73.20
LG1B/SG2B	AND	12.88	12.88	22.81
LG1B/SG2B	AVG	57.73	57.73	73.20
LG1B/WG7B	OR	68.45	68.45	81.27
LG1B/WG7B	AND	19.96	19.96	33.27
LG1B/WG7B	AVG	68.45	68.45	81.27
LG1B/LG8B	OR	68.45	68.45	81.27
LG1B/LG8B	AND	16.95	16.95	28.99
LG1B/LG8B	AVG	68.45	68.45	81.27
GG2B/SG2B	OR	51.50	51.50	67.99
GG2B/SG2B	AND	25.54	25.54	40.68
GG2B/SG2B	AVG	51.50	51.50	67.99
GG2B/WG7B	OR	57.94	57.94	73.37
GG2B/WG7B	AND	36.91	36.91	53.92
GG2B/WG7B	AVG	57.94	57.94	73.37
GG2B/LG8B	OR	64.59	64.59	78.49
GG2B/LG8B	AND	27.25	27.25	42.83
GG2B/LG8B	AVG	64.59	64.59	78.49
SG2B/WG7B	OR	49.79	49.79	66.48
SG2B/WG7B	AND	21.67	21.67	35.63
SG2B/WG7B	AVG	49.79	49.79	66.48
SG2B/LG8B	OR	58.37	58.37	73.71
SG2B/LG8B	AND	10.09	10.09	18.32
SG2B/LG8B	AVG	58.37	58.37	73.71
WG7B/LG8B	OR	61.80	61.80	76.39
WG7B/LG8B	AND	24.46	24.46	39.31
WG7B/LG8B	AVG	61.80	61.80	76.39
LG1B/GG2B/SG2B	OR	72.32	72.32	83.94
LG1B/GG2B/SG2B	AND	12.23	12.23	21.80
LG1B/GG2B/SG2B	AVG	36.27	36.27	53.23
LG1B/GG2B/WG7B	OR	76.18	76.18	86.48
LG1B/GG2B/WG7B	AND	16.74	16.74	28.68
LG1B/GG2B/WG7B	AVG	45.71	45.71	62.74
LG1B/GG2B/LG8B	OR	80.26	80.26	89.05
LG1B/GG2B/LG8B	AND	11.16	11.16	20.08
LG1B/GG2B/LG8B	AVG	44.21	44.21	61.31
LG1B/SG2B/WG7B	OR	71.03	71.03	83.06
LG1B/SG2B/WG7B	AND	10.30	10.30	18.68
LG1B/SG2B/WG7B	AVG	33.91	33.91	50.64
LG1B/SG2B/LG8B	OR	76.61	76.61	86.76
LG1B/SG2B/LG8B	AND	4.29	4.29	8.23
LG1B/SG2B/LG8B	AVG	31.33	31.33	47.71
LG1B/WG7B/LG8B	OR	79.18	79.18	88.38
LG1B/WG7B/LG8B	AND	10.52	10.52	19.03

Continued on next page

Table 106 – continued from previous page

Model	Logic	Accuracy(%)	Recall(%)	F1Score(%)
LG1B/WG7B/LG8B	AVG	40.34	40.34	57.49
GG2B/SG2B/WG7B	OR	58.58	58.58	73.88
GG2B/SG2B/WG7B	AND	21.03	21.03	34.75
GG2B/SG2B/WG7B	AVG	42.06	42.06	59.21
GG2B/SG2B/LG8B	OR	65.67	65.67	79.27
GG2B/SG2B/LG8B	AND	9.87	9.87	17.97
GG2B/SG2B/LG8B	AVG	43.13	43.13	60.27
GG2B/WG7B/LG8B	OR	69.31	69.31	81.88
GG2B/WG7B/LG8B	AND	21.46	21.46	35.34
GG2B/WG7B/LG8B	AVG	45.71	45.71	62.74
SG2B/WG7B/LG8B	OR	66.31	66.31	79.74
SG2B/WG7B/LG8B	AND	9.44	9.44	17.25
SG2B/WG7B/LG8B	AVG	37.34	37.34	54.37
LG1B/GG2B/SG2B/WG7B	OR	76.39	76.39	86.62
LG1B/GG2B/SG2B/WG7B	AND	10.09	10.09	18.32
LG1B/GG2B/SG2B/WG7B	AVG	48.93	48.93	65.71
LG1B/GG2B/SG2B/LG8B	OR	80.90	80.90	89.44
LG1B/GG2B/SG2B/LG8B	AND	4.08	4.08	7.84
LG1B/GG2B/SG2B/LG8B	AVG	52.15	52.15	68.55
LG1B/GG2B/WG7B/LG8B	OR	82.83	82.83	90.61
LG1B/GG2B/WG7B/LG8B	AND	9.44	9.44	17.25
LG1B/GG2B/WG7B/LG8B	AVG	56.44	56.44	72.15
LG1B/SG2B/WG7B/LG8B	OR	81.33	81.33	89.70
LG1B/SG2B/WG7B/LG8B	AND	4.08	4.08	7.84
LG1B/SG2B/WG7B/LG8B	AVG	49.14	49.14	65.90
GG2B/SG2B/WG7B/LG8B	OR	69.74	69.74	82.17
GG2B/SG2B/WG7B/LG8B	AND	9.44	9.44	17.25
GG2B/SG2B/WG7B/LG8B	AVG	50.64	50.64	67.24
LG1B/GG2B/SG2B/WG7B/LG8B	OR	**83.05**	**83.05**	**90.74**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	4.08	4.08	7.84
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	40.13	40.13	57.27

3.4 ExpGuardMix (Input only)

Table 107: ExpGuardMix (Input) - Table 1

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B	-	558	544	475	698	46.61	55.57
GG2B	-	1133	792	227	123	22.28	9.79
SG2B	-	178	1009	10	1078	0.98	85.83
WG7B	-	941	987	32	315	3.14	25.08
LG8B	-	742	922	97	514	9.52	40.92
LG1B/GG2B	OR	1180	424	595	76	58.39	6.05
LG1B/GG2B	AND	511	912	107	745	10.50	59.32

Continued on next page

Table 107 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B/GG2B	AVG	1180	424	595	76	58.39	6.05
LG1B/SG2B	OR	658	541	478	598	46.91	47.61
LG1B/SG2B	AND	78	1012	7	1178	0.69	93.79
LG1B/SG2B	AVG	658	541	478	598	46.91	47.61
LG1B/WG7B	OR	1077	528	491	179	48.18	14.25
LG1B/WG7B	AND	422	1003	16	834	1.57	66.40
LG1B/WG7B	AVG	1077	528	491	179	48.18	14.25
LG1B/LG8B	OR	965	495	524	291	51.42	23.17
LG1B/LG8B	AND	335	971	48	921	4.71	73.33
LG1B/LG8B	AVG	965	495	524	291	51.42	23.17
GG2B/SG2B	OR	1134	786	233	122	22.87	9.71
GG2B/SG2B	AND	177	1015	4	1079	0.39	85.91
GG2B/SG2B	AVG	1134	786	233	122	22.87	9.71
GG2B/WG7B	OR	1139	784	235	117	23.06	9.32
GG2B/WG7B	AND	935	995	24	321	2.36	25.56
GG2B/WG7B	AVG	1139	784	235	117	23.06	9.32
GG2B/LG8B	OR	1143	729	290	113	28.46	9.00
GG2B/LG8B	AND	732	985	34	524	3.34	41.72
GG2B/LG8B	AVG	1143	729	290	113	28.46	9.00
SG2B/WG7B	OR	950	981	38	306	3.73	24.36
SG2B/WG7B	AND	169	1015	4	1087	0.39	86.54
SG2B/WG7B	AVG	950	981	38	306	3.73	24.36
SG2B/LG8B	OR	769	913	106	487	10.40	38.77
SG2B/LG8B	AND	151	1018	1	1105	0.10	87.98
SG2B/LG8B	AVG	769	913	106	487	10.40	38.77
WG7B/LG8B	OR	985	898	121	271	11.87	21.58
WG7B/LG8B	AND	698	1011	8	558	0.79	44.43
WG7B/LG8B	AVG	985	898	121	271	11.87	21.58
LG1B/GG2B/SG2B	OR	1180	421	598	76	58.68	6.05
LG1B/GG2B/SG2B	AND	77	1015	4	1179	0.39	93.87
LG1B/GG2B/SG2B	AVG	612	909	110	644	10.79	51.27
LG1B/GG2B/WG7B	OR	1183	420	599	73	58.78	5.81
LG1B/GG2B/WG7B	AND	419	1007	12	837	1.18	66.64
LG1B/GG2B/WG7B	AVG	1030	896	123	226	12.07	17.99
LG1B/GG2B/LG8B	OR	1185	394	625	71	61.33	5.65
LG1B/GG2B/LG8B	AND	330	1004	15	926	1.47	73.73
LG1B/GG2B/LG8B	AVG	918	860	159	338	15.60	26.91
LG1B/SG2B/WG7B	OR	1079	527	492	177	48.28	14.09
LG1B/SG2B/WG7B	AND	71	1017	2	1185	0.20	94.35
LG1B/SG2B/WG7B	AVG	527	996	23	729	2.26	58.04
LG1B/SG2B/LG8B	OR	978	493	526	278	51.62	22.13
LG1B/SG2B/LG8B	AND	64	1019	0	1192	**0.00**	94.90
LG1B/SG2B/LG8B	AVG	436	963	56	820	5.50	65.29
LG1B/WG7B/LG8B	OR	1098	483	536	158	52.60	12.58
LG1B/WG7B/LG8B	AND	312	1015	4	944	0.39	75.16
LG1B/WG7B/LG8B	AVG	831	955	64	425	6.28	33.84

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Table 107 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
GG2B/SG2B/WG7B	OR	1140	781	238	116	23.36	9.24
GG2B/SG2B/WG7B	AND	169	1018	1	1087	0.10	86.54
GG2B/SG2B/WG7B	AVG	943	989	30	313	2.94	24.92
GG2B/SG2B/LG8B	OR	1144	724	295	112	28.95	8.92
GG2B/SG2B/LG8B	AND	151	1019	0	1105	**0.00**	87.98
GG2B/SG2B/LG8B	AVG	758	980	39	498	3.83	39.65
GG2B/WG7B/LG8B	OR	1148	723	296	108	29.05	8.60
GG2B/WG7B/LG8B	AND	697	1013	6	559	0.59	44.51
GG2B/WG7B/LG8B	AVG	971	965	54	285	5.30	22.69
SG2B/WG7B/LG8B	OR	990	892	127	266	12.46	21.18
SG2B/WG7B/LG8B	AND	147	1018	1	1109	0.10	88.30
SG2B/WG7B/LG8B	AVG	724	1008	11	532	1.08	42.36
LG1B/GG2B/SG2B/WG7B	OR	1183	419	600	73	58.88	5.81
LG1B/GG2B/SG2B/WG7B	AND	71	1018	1	1185	0.10	94.35
LG1B/GG2B/SG2B/WG7B	AVG	1033	892	127	223	12.46	17.75
LG1B/GG2B/SG2B/LG8B	OR	1185	392	627	71	61.53	5.65
LG1B/GG2B/SG2B/LG8B	AND	64	1019	0	1192	**0.00**	94.90
LG1B/GG2B/SG2B/LG8B	AVG	932	856	163	324	16.00	25.80
LG1B/GG2B/WG7B/LG8B	OR	1187	392	627	69	61.53	**5.49**
LG1B/GG2B/WG7B/LG8B	AND	312	1015	4	944	0.39	75.16
LG1B/GG2B/WG7B/LG8B	AVG	1053	844	175	203	17.17	16.16
LG1B/SG2B/WG7B/LG8B	OR	1098	482	537	158	52.70	12.58
LG1B/SG2B/WG7B/LG8B	AND	62	1019	0	1194	**0.00**	95.06
LG1B/SG2B/WG7B/LG8B	AVG	851	949	70	405	6.87	32.25
GG2B/SG2B/WG7B/LG8B	OR	1149	720	299	107	29.34	8.52
GG2B/SG2B/WG7B/LG8B	AND	147	1019	0	1109	**0.00**	88.30
GG2B/SG2B/WG7B/LG8B	AVG	975	960	59	281	5.79	22.37
LG1B/GG2B/SG2B/WG7B/LG8B	OR	1187	391	628	69	61.63	**5.49**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	62	1019	0	1194	**0.00**	95.06
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	841	989	30	415	2.94	33.04

Table 108: ExpGuardMix (Input) - Table 2

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	-	48.44	53.39	54.02	44.43	48.75	-2.18
GG2B	-	84.62	77.72	83.31	90.21	**86.62**	68.89
SG2B	-	52.18	99.02	94.68	14.17	24.65	23.82
WG7B	-	84.75	96.86	96.71	74.92	84.43	72.15
LG8B	-	73.14	90.48	88.44	59.08	70.84	51.08
LG1B/GG2B	OR	70.51	41.61	66.48	93.95	77.86	42.70
LG1B/GG2B	AND	62.55	89.50	82.69	40.68	54.54	33.74
LG1B/GG2B	AVG	70.51	41.61	66.48	93.95	77.86	42.70
LG1B/SG2B	OR	52.70	53.09	57.92	52.39	55.02	5.45
LG1B/SG2B	AND	47.91	99.31	91.76	6.21	11.63	14.48

Continued on next page

Table 108 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B/SG2B	AVG	52.70	53.09	57.92	52.39	55.02	5.45
LG1B/WG7B	OR	70.55	51.82	68.69	85.75	76.27	40.36
LG1B/WG7B	AND	62.64	98.43	96.35	33.60	49.82	40.40
LG1B/WG7B	AVG	70.55	51.82	68.69	85.75	76.27	40.36
LG1B/LG8B	OR	64.18	48.58	64.81	76.83	70.31	26.57
LG1B/LG8B	AND	57.41	95.29	87.47	26.67	40.88	29.19
LG1B/LG8B	AVG	64.18	48.58	64.81	76.83	70.31	26.57
GG2B/SG2B	OR	84.40	77.13	82.96	90.29	86.47	68.46
GG2B/SG2B	AND	52.40	99.61	97.79	14.09	24.63	25.17
GG2B/SG2B	AVG	84.40	77.13	82.96	90.29	86.47	68.46
GG2B/WG7B	OR	84.53	76.94	82.90	90.68	86.62	68.76
GG2B/WG7B	AND	84.84	97.64	97.50	74.44	84.42	**72.59**
GG2B/WG7B	AVG	84.53	76.94	82.90	90.68	86.62	68.76
GG2B/LG8B	OR	82.29	71.54	79.76	91.00	85.01	64.42
GG2B/LG8B	AND	75.47	96.66	95.56	58.28	72.40	57.81
GG2B/LG8B	AVG	82.29	71.54	79.76	91.00	85.01	64.42
SG2B/WG7B	OR	84.88	96.27	96.15	75.64	84.67	72.14
SG2B/WG7B	AND	52.04	99.61	97.69	13.46	23.65	24.51
SG2B/WG7B	AVG	84.88	96.27	96.15	75.64	84.67	72.14
SG2B/LG8B	OR	73.93	89.60	87.89	61.23	72.17	51.95
SG2B/LG8B	AND	51.38	99.90	99.34	12.02	21.45	23.75
SG2B/LG8B	AVG	73.93	89.60	87.89	61.23	72.17	51.95
WG7B/LG8B	OR	82.77	88.13	89.06	78.42	83.40	66.21
WG7B/LG8B	AND	75.12	99.21	98.87	55.57	71.15	58.89
WG7B/LG8B	AVG	82.77	88.13	89.06	78.42	83.40	66.21
LG1B/GG2B/SG2B	OR	70.37	41.32	66.37	93.95	77.79	42.44
LG1B/GG2B/SG2B	AND	48.00	99.61	95.06	6.13	11.52	15.40
LG1B/GG2B/SG2B	AVG	66.86	89.21	84.76	48.73	61.88	40.53
LG1B/GG2B/WG7B	OR	70.46	41.22	66.39	94.19	77.88	42.73
LG1B/GG2B/WG7B	AND	62.68	98.82	97.22	33.36	49.67	40.84
LG1B/GG2B/WG7B	AVG	84.66	87.93	89.33	82.01	85.51	69.56
LG1B/GG2B/LG8B	OR	69.41	38.67	65.47	94.35	77.30	40.71
LG1B/GG2B/LG8B	AND	58.64	98.53	95.65	26.27	41.22	34.39
LG1B/GG2B/LG8B	AVG	78.15	84.40	85.24	73.09	78.70	57.25
LG1B/SG2B/WG7B	OR	70.59	51.72	68.68	85.91	76.34	40.47
LG1B/SG2B/WG7B	AND	47.82	99.80	97.26	5.65	10.68	15.40
LG1B/SG2B/WG7B	AVG	66.95	97.74	95.82	41.96	58.36	46.11
LG1B/SG2B/LG8B	OR	64.66	48.38	65.03	77.87	70.87	27.57
LG1B/SG2B/LG8B	AND	47.60	**100.00**	**100.00**	5.10	9.70	15.32
LG1B/SG2B/LG8B	AVG	61.49	94.50	88.62	34.71	49.89	35.29
LG1B/WG7B/LG8B	OR	69.49	47.40	67.20	87.42	75.99	38.49
LG1B/WG7B/LG8B	AND	58.33	99.61	98.73	24.84	39.69	35.15
LG1B/WG7B/LG8B	AVG	78.51	93.72	92.85	66.16	77.27	60.96
GG2B/SG2B/WG7B	OR	84.44	76.64	82.73	90.76	86.56	68.59
GG2B/SG2B/WG7B	AND	52.18	99.90	99.41	13.46	23.70	25.26
GG2B/SG2B/WG7B	AVG	84.92	97.06	96.92	75.08	84.61	72.51

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Table 108 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
GG2B/SG2B/LG8B	OR	82.11	71.05	79.50	91.08	84.90	64.09
GG2B/SG2B/LG8B	AND	51.43	**100.00**	**100.00**	12.02	21.46	24.02
GG2B/SG2B/LG8B	AVG	76.40	96.17	95.11	60.35	73.84	58.92
GG2B/WG7B/LG8B	OR	82.24	70.95	79.50	91.40	85.04	64.40
GG2B/WG7B/LG8B	AND	75.16	99.41	99.15	55.49	71.16	59.09
GG2B/WG7B/LG8B	AVG	**85.10**	94.70	94.73	77.31	85.14	71.97
SG2B/WG7B/LG8B	OR	82.73	87.54	88.63	78.82	83.44	66.01
SG2B/WG7B/LG8B	AND	51.21	99.90	99.32	11.70	20.94	23.40
SG2B/WG7B/LG8B	AVG	76.13	98.92	98.50	57.64	72.73	60.15
LG1B/GG2B/SG2B/WG7B	OR	70.42	41.12	66.35	94.19	77.85	42.65
LG1B/GG2B/SG2B/WG7B	AND	47.87	99.90	98.61	5.65	10.69	15.78
LG1B/GG2B/SG2B/WG7B	AVG	84.62	87.54	89.05	82.25	85.51	69.42
LG1B/GG2B/SG2B/LG8B	OR	69.32	38.47	65.40	94.35	77.25	40.53
LG1B/GG2B/SG2B/LG8B	AND	47.60	**100.00**	**100.00**	5.10	9.70	15.32
LG1B/GG2B/SG2B/LG8B	AVG	78.59	84.00	85.11	74.20	79.29	57.93
LG1B/GG2B/WG7B/LG8B	OR	69.41	38.47	65.44	**94.51**	77.33	40.79
LG1B/GG2B/WG7B/LG8B	AND	58.33	99.61	98.73	24.84	39.69	35.15
LG1B/GG2B/WG7B/LG8B	AVG	83.38	82.83	85.75	83.84	84.78	66.51
LG1B/SG2B/WG7B/LG8B	OR	69.45	47.30	67.16	87.42	75.96	38.40
LG1B/SG2B/WG7B/LG8B	AND	47.52	**100.00**	**100.00**	4.94	9.41	15.08
LG1B/SG2B/WG7B/LG8B	AVG	79.12	93.13	92.40	67.75	78.18	61.68
GG2B/SG2B/WG7B/LG8B	OR	82.15	70.66	79.35	91.48	84.99	64.24
GG2B/SG2B/WG7B/LG8B	AND	51.25	**100.00**	**100.00**	11.70	20.96	23.67
GG2B/SG2B/WG7B/LG8B	AVG	85.05	94.21	94.29	77.63	85.15	71.74
LG1B/GG2B/SG2B/WG7B/LG8B	OR	69.36	38.37	65.40	**94.51**	77.30	40.71
LG1B/GG2B/SG2B/WG7B/LG8B	AND	47.52	**100.00**	**100.00**	4.94	9.41	15.08
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	80.44	97.06	96.56	66.96	79.08	65.49

3.5 ExpGuardMix (Input & Output)

Table 109: ExpGuardMix (Input&Output) - Table 1

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B	-	621	523	536	595	50.61	48.93
GG2B	-	1035	853	206	181	19.45	14.88
SG2B	-	97	1053	6	1119	0.57	92.02
WG7B	-	716	1034	25	500	2.36	41.12
LG8B	-	729	962	97	487	9.16	40.05
LG1B/GG2B	OR	1120	414	645	96	60.91	7.89
LG1B/GG2B	AND	536	962	97	680	9.16	55.92
LG1B/GG2B	AVG	1120	414	645	96	60.91	7.89
LG1B/SG2B	OR	668	520	539	548	50.90	45.07
LG1B/SG2B	AND	50	1056	3	1166	0.28	95.89
LG1B/SG2B	AVG	668	520	539	548	50.90	45.07

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Table 109 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B/WG7B	OR	960	510	549	256	51.84	21.05
LG1B/WG7B	AND	377	1047	12	839	1.13	69.00
LG1B/WG7B	AVG	960	510	549	256	51.84	21.05
LG1B/LG8B	OR	973	475	584	243	55.15	19.98
LG1B/LG8B	AND	377	1010	49	839	4.63	69.00
LG1B/LG8B	AVG	973	475	584	243	55.15	19.98
GG2B/SG2B	OR	1038	849	210	178	19.83	14.64
GG2B/SG2B	AND	94	1057	2	1122	0.19	92.27
GG2B/SG2B	AVG	1038	849	210	178	19.83	14.64
GG2B/WG7B	OR	1048	846	213	168	20.11	13.82
GG2B/WG7B	AND	703	1041	18	513	1.70	42.19
GG2B/WG7B	AVG	1048	846	213	168	20.11	13.82
GG2B/LG8B	OR	1072	797	262	144	24.74	11.84
GG2B/LG8B	AND	692	1018	41	524	3.87	43.09
GG2B/LG8B	AVG	1072	797	262	144	24.74	11.84
SG2B/WG7B	OR	724	1032	27	492	2.55	40.46
SG2B/WG7B	AND	89	1055	4	1127	0.38	92.68
SG2B/WG7B	AVG	724	1032	27	492	2.55	40.46
SG2B/LG8B	OR	743	957	102	473	9.63	38.90
SG2B/LG8B	AND	83	1058	1	1133	0.09	93.17
SG2B/LG8B	AVG	743	957	102	473	9.63	38.90
WG7B/LG8B	OR	885	945	114	331	10.76	27.22
WG7B/LG8B	AND	560	1051	8	656	0.76	53.95
WG7B/LG8B	AVG	885	945	114	331	10.76	27.22
LG1B/GG2B/SG2B	OR	1121	413	646	95	61.00	7.81
LG1B/GG2B/SG2B	AND	48	1059	0	1168	**0.00**	96.05
LG1B/GG2B/SG2B	AVG	584	957	102	632	9.63	51.97
LG1B/GG2B/WG7B	OR	1126	411	648	90	61.19	7.40
LG1B/GG2B/WG7B	AND	370	1051	8	846	0.76	69.57
LG1B/GG2B/WG7B	AVG	876	948	111	340	10.48	27.96
LG1B/GG2B/LG8B	OR	1143	389	670	73	63.27	6.00
LG1B/GG2B/LG8B	AND	363	1041	18	853	1.70	70.15
LG1B/GG2B/LG8B	AVG	879	908	151	337	14.26	27.71
LG1B/SG2B/WG7B	OR	965	509	550	251	51.94	20.64
LG1B/SG2B/WG7B	AND	47	1057	2	1169	0.19	96.13
LG1B/SG2B/WG7B	AVG	422	1044	15	794	1.42	65.30
LG1B/SG2B/LG8B	OR	980	473	586	236	55.34	19.41
LG1B/SG2B/LG8B	AND	43	1059	0	1173	**0.00**	96.46
LG1B/SG2B/LG8B	AVG	424	1006	53	792	5.00	65.13
LG1B/WG7B/LG8B	OR	1044	469	590	172	55.71	14.14
LG1B/WG7B/LG8B	AND	292	1058	1	924	0.09	75.99
LG1B/WG7B/LG8B	AVG	730	992	67	486	6.33	39.97
GG2B/SG2B/WG7B	OR	1049	845	214	167	20.21	13.73
GG2B/SG2B/WG7B	AND	87	1058	1	1129	0.09	92.85
GG2B/SG2B/WG7B	AVG	712	1037	22	504	2.08	41.45
GG2B/SG2B/LG8B	OR	1075	794	265	141	25.02	11.60

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Table 109 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
GG2B/SG2B/LG8B	AND	83	1059	0	1133	**0.00**	93.17
GG2B/SG2B/LG8B	AVG	703	1015	44	513	4.15	42.19
GG2B/WG7B/LG8B	OR	1079	791	268	137	25.31	11.27
GG2B/WG7B/LG8B	AND	554	1052	7	662	0.66	54.44
GG2B/WG7B/LG8B	AVG	847	1006	53	369	5.00	30.35
SG2B/WG7B/LG8B	OR	888	943	116	328	10.95	26.97
SG2B/WG7B/LG8B	AND	78	1058	1	1138	0.09	93.59
SG2B/WG7B/LG8B	AVG	576	1048	11	640	1.04	52.63
LG1B/GG2B/SG2B/WG7B	OR	1127	411	648	89	61.19	7.32
LG1B/GG2B/SG2B/WG7B	AND	45	1059	0	1171	**0.00**	96.30
LG1B/GG2B/SG2B/WG7B	AVG	880	945	114	336	10.76	27.63
LG1B/GG2B/SG2B/LG8B	OR	1144	389	670	72	63.27	5.92
LG1B/GG2B/SG2B/LG8B	AND	43	1059	0	1173	**0.00**	96.46
LG1B/GG2B/SG2B/LG8B	AVG	887	902	157	329	14.83	27.06
LG1B/GG2B/WG7B/LG8B	OR	1144	387	672	72	63.46	5.92
LG1B/GG2B/WG7B/LG8B	AND	291	1058	1	925	0.09	76.07
LG1B/GG2B/WG7B/LG8B	AVG	960	899	160	256	15.11	21.05
LG1B/SG2B/WG7B/LG8B	OR	1047	468	591	169	55.81	13.90
LG1B/SG2B/WG7B/LG8B	AND	40	1059	0	1176	**0.00**	96.71
LG1B/SG2B/WG7B/LG8B	AVG	736	990	69	480	6.52	39.47
GG2B/SG2B/WG7B/LG8B	OR	1080	790	269	136	25.40	11.18
GG2B/SG2B/WG7B/LG8B	AND	78	1059	0	1138	**0.00**	93.59
GG2B/SG2B/WG7B/LG8B	AVG	851	1003	56	365	5.29	30.02
LG1B/GG2B/SG2B/WG7B/LG8B	OR	1145	387	672	71	63.46	**5.84**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	40	1059	0	1176	**0.00**	96.71
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	714	1024	35	502	3.31	41.28

Table 110: ExpGuardMix (Input&Output) - Table 2

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	-	50.29	49.39	53.67	51.07	52.34	0.45
GG2B	-	82.99	80.55	83.40	85.12	84.25	65.78
SG2B	-	50.55	99.43	94.17	7.98	14.71	17.78
WG7B	-	76.92	97.64	96.63	58.88	73.17	60.16
LG8B	-	74.33	90.84	88.26	59.95	71.40	52.68
LG1B/GG2B	OR	67.43	39.09	63.46	92.11	75.14	37.32
LG1B/GG2B	AND	65.85	90.84	84.68	44.08	57.98	38.87
LG1B/GG2B	AVG	67.43	39.09	63.46	92.11	75.14	37.32
LG1B/SG2B	OR	52.22	49.10	55.34	54.93	55.14	4.04
LG1B/SG2B	AND	48.62	99.72	94.34	4.11	7.88	12.66
LG1B/SG2B	AVG	52.22	49.10	55.34	54.93	55.14	4.04
LG1B/WG7B	OR	64.62	48.16	63.62	78.95	70.46	28.61
LG1B/WG7B	AND	62.59	98.87	96.92	31.00	46.98	39.57
LG1B/WG7B	AVG	64.62	48.16	63.62	78.95	70.46	28.61

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Table 110 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B/LG8B	OR	63.65	44.85	62.49	80.02	70.18	26.69
LG1B/LG8B	AND	60.97	95.37	88.50	31.00	45.92	33.73
LG1B/LG8B	AVG	63.65	44.85	62.49	80.02	70.18	26.69
GG2B/SG2B	OR	82.95	80.17	83.17	85.36	84.25	65.69
GG2B/SG2B	AND	50.59	99.81	97.92	7.73	14.33	18.71
GG2B/SG2B	AVG	82.95	80.17	83.17	85.36	84.25	65.69
GG2B/WG7B	OR	**83.25**	79.89	83.11	86.18	84.62	66.31
GG2B/WG7B	AND	76.66	98.30	97.50	57.81	72.59	60.16
GG2B/WG7B	AVG	**83.25**	79.89	83.11	86.18	84.62	66.31
GG2B/LG8B	OR	82.15	75.26	80.36	88.16	84.08	64.23
GG2B/LG8B	AND	75.16	96.13	94.41	56.91	71.01	56.61
GG2B/LG8B	AVG	82.15	75.26	80.36	88.16	84.08	64.23
SG2B/WG7B	OR	77.19	97.45	96.40	59.54	73.61	60.45
SG2B/WG7B	AND	50.29	99.62	95.70	7.32	13.60	17.49
SG2B/WG7B	AVG	77.19	97.45	96.40	59.54	73.61	60.45
SG2B/LG8B	OR	74.73	90.37	87.93	61.10	72.10	53.13
SG2B/LG8B	AND	50.15	99.91	98.81	6.83	12.77	17.81
SG2B/LG8B	AVG	74.73	90.37	87.93	61.10	72.10	53.13
WG7B/LG8B	OR	80.44	89.24	88.59	72.78	79.91	62.33
WG7B/LG8B	AND	70.81	99.24	98.59	46.05	62.78	52.20
WG7B/LG8B	AVG	80.44	89.24	88.59	72.78	79.91	62.33
LG1B/GG2B/SG2B	OR	67.43	39.00	63.44	92.19	75.16	37.35
LG1B/GG2B/SG2B	AND	48.66	**100.00**	**100.00**	3.95	7.59	13.70
LG1B/GG2B/SG2B	AVG	67.74	90.37	85.13	48.03	61.41	41.73
LG1B/GG2B/WG7B	OR	67.56	38.81	63.47	92.60	75.32	37.81
LG1B/GG2B/WG7B	AND	62.46	99.24	97.88	30.43	46.42	39.76
LG1B/GG2B/WG7B	AVG	80.18	89.52	88.75	72.04	79.53	61.96
LG1B/GG2B/LG8B	OR	67.34	36.73	63.04	94.00	75.47	38.10
LG1B/GG2B/LG8B	AND	61.71	98.30	95.28	29.85	45.46	37.61
LG1B/GG2B/LG8B	AVG	78.55	85.74	85.34	72.29	78.27	58.15
LG1B/SG2B/WG7B	OR	64.79	48.06	63.70	79.36	70.67	29.00
LG1B/SG2B/WG7B	AND	48.53	99.81	95.92	3.87	7.43	12.63
LG1B/SG2B/WG7B	AVG	64.44	98.58	96.57	34.70	51.06	42.15
LG1B/SG2B/LG8B	OR	63.87	44.66	62.58	80.59	70.45	27.20
LG1B/SG2B/LG8B	AND	48.44	**100.00**	**100.00**	3.54	6.83	12.95
LG1B/SG2B/LG8B	AVG	62.86	95.00	88.89	34.87	50.09	36.59
LG1B/WG7B/LG8B	OR	66.51	44.29	63.89	85.86	73.26	33.42
LG1B/WG7B/LG8B	AND	59.34	99.91	99.66	24.01	38.70	35.62
LG1B/WG7B/LG8B	AVG	75.69	93.67	91.59	60.03	72.53	56.15
GG2B/SG2B/WG7B	OR	**83.25**	79.79	83.06	86.27	**84.63**	**66.31**
GG2B/SG2B/WG7B	AND	50.33	99.91	98.86	7.15	13.34	18.26
GG2B/SG2B/WG7B	AVG	76.88	97.92	97.00	58.55	73.03	60.26
GG2B/SG2B/LG8B	OR	82.15	74.98	80.22	88.40	84.12	64.26
GG2B/SG2B/LG8B	AND	50.20	**100.00**	**100.00**	6.83	12.78	18.16
GG2B/SG2B/LG8B	AVG	75.52	95.85	94.11	57.81	71.63	56.99
GG2B/WG7B/LG8B	OR	82.20	74.69	80.10	88.73	84.20	64.38

Continued on next page

Table 110 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
GG2B/WG7B/LG8B	AND	70.59	99.34	98.75	45.56	62.35	51.96
GG2B/WG7B/LG8B	AVG	81.45	95.00	94.11	69.65	80.06	65.95
SG2B/WG7B/LG8B	OR	80.48	89.05	88.45	73.03	80.00	62.36
SG2B/WG7B/LG8B	AND	49.93	99.91	98.73	6.41	12.05	17.22
SG2B/WG7B/LG8B	AVG	71.38	98.96	98.13	47.37	63.89	52.82
LG1B/GG2B/SG2B/WG7B	OR	67.60	38.81	63.49	92.68	75.36	37.93
LG1B/GG2B/SG2B/WG7B	AND	48.53	**100.00**	**100.00**	3.70	7.14	13.26
LG1B/GG2B/SG2B/WG7B	AVG	80.22	89.24	88.53	72.37	79.64	61.95
LG1B/GG2B/SG2B/LG8B	OR	67.38	36.73	63.07	94.08	75.51	38.24
LG1B/GG2B/SG2B/LG8B	AND	48.44	**100.00**	**100.00**	3.54	6.83	12.95
LG1B/GG2B/SG2B/LG8B	AVG	78.64	85.17	84.96	72.94	78.50	58.18
LG1B/GG2B/WG7B/LG8B	OR	67.30	36.54	63.00	94.08	75.46	38.06
LG1B/GG2B/WG7B/LG8B	AND	59.30	99.91	99.66	23.93	38.59	35.55
LG1B/GG2B/WG7B/LG8B	AVG	81.71	84.89	85.71	78.95	82.19	63.69
LG1B/SG2B/WG7B/LG8B	OR	66.59	44.19	63.92	86.10	73.37	33.66
LG1B/SG2B/WG7B/LG8B	AND	48.31	**100.00**	**100.00**	3.29	6.37	12.48
LG1B/SG2B/WG7B/LG8B	AVG	75.87	93.48	91.43	60.53	72.84	56.34
GG2B/SG2B/WG7B/LG8B	OR	82.20	74.60	80.06	88.82	84.21	64.39
GG2B/SG2B/WG7B/LG8B	AND	49.98	**100.00**	**100.00**	6.41	12.06	17.58
GG2B/SG2B/WG7B/LG8B	AVG	81.49	94.71	93.83	69.98	80.17	65.91
LG1B/GG2B/SG2B/WG7B/LG8B	OR	67.34	36.54	63.02	**94.16**	75.50	38.20
LG1B/GG2B/SG2B/WG7B/LG8B	AND	48.31	**100.00**	**100.00**	3.29	6.37	12.48
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	76.40	96.69	95.33	58.72	72.67	58.82

3.6 WildGuardMix

Table 111: WildGuardMix - Table 1

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B	-	160	621	804	140	56.42	46.67
GG2B	-	224	1347	78	76	5.47	25.33
SG2B	-	99	1401	24	201	1.68	67.00
WG7B	-	196	1396	29	104	2.04	34.67
LG8B	-	180	1388	37	120	2.60	40.00
LG1B/GG2B	OR	267	588	837	33	58.74	11.00
LG1B/GG2B	AND	117	1380	45	183	3.16	61.00
LG1B/GG2B	AVG	267	588	837	33	58.74	11.00
LG1B/SG2B	OR	204	609	816	96	57.26	32.00
LG1B/SG2B	AND	55	1413	12	245	0.84	81.67
LG1B/SG2B	AVG	204	609	816	96	57.26	32.00
LG1B/WG7B	OR	257	611	814	43	57.12	14.33
LG1B/WG7B	AND	99	1406	19	201	1.33	67.00
LG1B/WG7B	AVG	257	611	814	43	57.12	14.33
LG1B/LG8B	OR	240	608	817	60	57.33	20.00

Continued on next page

Table 111 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
LG1B/LG8B	AND	100	1401	24	200	1.68	66.67
LG1B/LG8B	AVG	240	608	817	60	57.33	20.00
GG2B/SG2B	OR	232	1338	87	68	6.11	22.67
GG2B/SG2B	AND	91	1410	15	209	1.05	69.67
GG2B/SG2B	AVG	232	1338	87	68	6.11	22.67
GG2B/WG7B	OR	238	1334	91	62	6.39	20.67
GG2B/WG7B	AND	182	1409	16	118	1.12	39.33
GG2B/WG7B	AVG	238	1334	91	62	6.39	20.67
GG2B/LG8B	OR	241	1326	99	59	6.95	19.67
GG2B/LG8B	AND	163	1409	16	137	1.12	45.67
GG2B/LG8B	AVG	241	1326	99	59	6.95	19.67
SG2B/WG7B	OR	207	1378	47	93	3.30	31.00
SG2B/WG7B	AND	88	1419	6	212	0.42	70.67
SG2B/WG7B	AVG	207	1378	47	93	3.30	31.00
SG2B/LG8B	OR	195	1369	56	105	3.93	35.00
SG2B/LG8B	AND	84	1420	5	216	0.35	72.00
SG2B/LG8B	AVG	195	1369	56	105	3.93	35.00
WG7B/LG8B	OR	223	1373	52	77	3.65	25.67
WG7B/LG8B	AND	153	1411	14	147	0.98	49.00
WG7B/LG8B	AVG	223	1373	52	77	3.65	25.67
LG1B/GG2B/SG2B	OR	270	582	843	30	59.16	10.00
LG1B/GG2B/SG2B	AND	50	1416	9	250	0.63	83.33
LG1B/GG2B/SG2B	AVG	163	1371	54	137	3.79	45.67
LG1B/GG2B/WG7B	OR	273	582	843	27	59.16	9.00
LG1B/GG2B/WG7B	AND	91	1413	12	209	0.84	69.67
LG1B/GG2B/WG7B	AVG	216	1369	56	84	3.93	28.00
LG1B/GG2B/LG8B	OR	274	579	846	26	59.37	8.67
LG1B/GG2B/LG8B	AND	90	1413	12	210	0.84	70.00
LG1B/GG2B/LG8B	AVG	200	1364	61	100	4.28	33.33
LG1B/SG2B/WG7B	OR	261	601	824	39	57.82	13.00
LG1B/SG2B/WG7B	AND	48	1421	4	252	0.28	84.00
LG1B/SG2B/WG7B	AVG	146	1396	29	154	2.04	51.33
LG1B/SG2B/LG8B	OR	248	597	828	52	58.11	17.33
LG1B/SG2B/LG8B	AND	48	1421	4	252	0.28	84.00
LG1B/SG2B/LG8B	AVG	143	1392	33	157	2.32	52.33
LG1B/WG7B/LG8B	OR	265	603	822	35	57.68	11.67
LG1B/WG7B/LG8B	AND	81	1416	9	219	0.63	73.00
LG1B/WG7B/LG8B	AVG	190	1386	39	110	2.74	36.67
GG2B/SG2B/WG7B	OR	244	1325	100	56	7.02	18.67
GG2B/SG2B/WG7B	AND	86	1419	6	214	0.42	71.33
GG2B/SG2B/WG7B	AVG	189	1400	25	111	1.75	37.00
GG2B/SG2B/LG8B	OR	245	1318	107	55	7.51	18.33
GG2B/SG2B/LG8B	AND	80	1421	4	220	0.28	73.33
GG2B/SG2B/LG8B	AVG	178	1397	28	122	1.96	40.67
GG2B/WG7B/LG8B	OR	246	1322	103	54	7.23	18.00
GG2B/WG7B/LG8B	AND	144	1420	5	156	0.35	52.00

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Table 111 – continued from previous page

Model	Logic	TP	TN	FP	FN	FPR(%)	FNR(%)
GG2B/WG7B/LG8B	AVG	210	1389	36	90	2.53	30.00
SG2B/WG7B/LG8B	OR	230	1360	65	70	4.56	23.33
SG2B/WG7B/LG8B	AND	80	1425	0	220	**0.00**	73.33
SG2B/WG7B/LG8B	AVG	165	1400	25	135	1.75	45.00
LG1B/GG2B/SG2B/WG7B	OR	275	576	849	25	59.58	8.33
LG1B/GG2B/SG2B/WG7B	AND	47	1421	4	253	0.28	84.33
LG1B/GG2B/SG2B/WG7B	AVG	223	1362	63	77	4.42	25.67
LG1B/GG2B/SG2B/LG8B	OR	276	574	851	24	59.72	8.00
LG1B/GG2B/SG2B/LG8B	AND	45	1421	4	255	0.28	85.00
LG1B/GG2B/SG2B/LG8B	AVG	209	1354	71	91	4.98	30.33
LG1B/GG2B/WG7B/LG8B	OR	275	577	848	25	59.51	8.33
LG1B/GG2B/WG7B/LG8B	AND	77	1421	4	223	0.28	74.33
LG1B/GG2B/WG7B/LG8B	AVG	233	1355	70	67	4.91	22.33
LG1B/SG2B/WG7B/LG8B	OR	269	594	831	31	58.32	10.33
LG1B/SG2B/WG7B/LG8B	AND	44	1425	0	256	**0.00**	85.33
LG1B/SG2B/WG7B/LG8B	AVG	197	1379	46	103	3.23	34.33
GG2B/SG2B/WG7B/LG8B	OR	250	1314	111	50	7.79	16.67
GG2B/SG2B/WG7B/LG8B	AND	78	1425	0	222	**0.00**	74.00
GG2B/SG2B/WG7B/LG8B	AVG	215	1383	42	85	2.95	28.33
LG1B/GG2B/SG2B/WG7B/LG8B	OR	277	572	853	23	59.86	**7.67**
LG1B/GG2B/SG2B/WG7B/LG8B	AND	43	1425	0	257	**0.00**	85.67
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	183	1396	29	117	2.04	39.00

Table 112: WildGuardMix - Table 2

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
LG1B	-	45.28	43.58	16.60	53.33	25.32	-2.36
GG2B	-	91.07	94.53	74.17	74.67	74.42	69.01
SG2B	-	86.96	98.32	80.49	33.00	46.81	46.13
WG7B	-	92.29	97.96	87.11	65.33	74.67	71.24
LG8B	-	90.90	97.40	82.95	60.00	69.63	65.61
LG1B/GG2B	OR	49.57	41.26	24.18	89.00	38.03	23.90
LG1B/GG2B	AND	86.78	96.84	72.22	39.00	50.65	46.57
LG1B/GG2B	AVG	49.57	41.26	24.18	89.00	38.03	23.90
LG1B/SG2B	OR	47.13	42.74	20.00	68.00	30.91	8.28
LG1B/SG2B	AND	85.10	99.16	82.09	18.33	29.97	34.31
LG1B/SG2B	AVG	47.13	42.74	20.00	68.00	30.91	8.28
LG1B/WG7B	OR	50.32	42.88	24.00	85.67	37.49	22.30
LG1B/WG7B	AND	87.25	98.67	83.90	33.00	47.37	47.55
LG1B/WG7B	AVG	50.32	42.88	24.00	85.67	37.49	22.30
LG1B/LG8B	OR	49.16	42.67	22.71	80.00	35.37	17.64
LG1B/LG8B	AND	87.01	98.32	80.65	33.33	47.17	46.44
LG1B/LG8B	AVG	49.16	42.67	22.71	80.00	35.37	17.64
GG2B/SG2B	OR	91.01	93.89	72.73	77.33	74.96	69.54

Continued on next page

Table 112 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
GG2B/SG2B	AND	87.01	98.95	85.85	30.33	44.83	46.21
GG2B/SG2B	AVG	91.01	93.89	72.73	77.33	74.96	69.54
GG2B/WG7B	OR	91.13	93.61	72.34	79.33	75.68	70.38
GG2B/WG7B	AND	92.23	98.88	91.92	60.67	73.09	70.80
GG2B/WG7B	AVG	91.13	93.61	72.34	79.33	75.68	70.38
GG2B/LG8B	OR	90.84	93.05	70.88	80.33	75.31	69.92
GG2B/LG8B	AND	91.13	98.88	91.06	54.33	68.06	66.14
GG2B/LG8B	AVG	90.84	93.05	70.88	80.33	75.31	69.92
SG2B/WG7B	OR	91.88	96.70	81.50	69.00	74.73	70.28
SG2B/WG7B	AND	87.36	99.58	93.62	29.33	44.67	48.28
SG2B/WG7B	AVG	91.88	96.70	81.50	69.00	74.73	70.28
SG2B/LG8B	OR	90.67	96.07	77.69	65.00	70.78	65.65
SG2B/LG8B	AND	87.19	99.65	94.38	28.00	43.19	47.38
SG2B/LG8B	AVG	90.67	96.07	77.69	65.00	70.78	65.65
WG7B/LG8B	OR	92.52	96.35	81.09	74.33	**77.57**	**73.19**
WG7B/LG8B	AND	90.67	99.02	91.62	51.00	65.52	64.11
WG7B/LG8B	AVG	92.52	96.35	81.09	74.33	**77.57**	**73.19**
LG1B/GG2B/SG2B	OR	49.39	40.84	24.26	90.00	38.22	24.43
LG1B/GG2B/SG2B	AND	84.99	99.37	84.75	16.67	27.86	33.44
LG1B/GG2B/SG2B	AVG	88.93	96.21	75.12	54.33	63.06	57.77
LG1B/GG2B/WG7B	OR	49.57	40.84	24.46	91.00	38.56	25.25
LG1B/GG2B/WG7B	AND	87.19	99.16	88.35	30.33	45.16	47.18
LG1B/GG2B/WG7B	AVG	91.88	96.07	79.41	72.00	75.52	70.80
LG1B/GG2B/LG8B	OR	49.45	40.63	24.46	91.33	38.59	25.39
LG1B/GG2B/LG8B	AND	87.13	99.16	88.24	30.00	44.78	46.86
LG1B/GG2B/LG8B	AVG	90.67	95.72	76.63	66.67	71.30	65.99
LG1B/SG2B/WG7B	OR	49.97	42.18	24.06	87.00	37.69	22.89
LG1B/SG2B/WG7B	AND	85.16	99.72	92.31	16.00	27.27	34.85
LG1B/SG2B/WG7B	AVG	89.39	97.96	83.43	48.67	61.47	58.54
LG1B/SG2B/LG8B	OR	48.99	41.89	23.05	82.67	36.05	19.22
LG1B/SG2B/LG8B	AND	85.16	99.72	92.31	16.00	27.27	34.85
LG1B/SG2B/LG8B	AVG	88.99	97.68	81.25	47.67	60.08	56.79
LG1B/WG7B/LG8B	OR	50.32	42.32	24.38	88.33	38.21	24.06
LG1B/WG7B/LG8B	AND	86.78	99.37	90.00	27.00	41.54	44.94
LG1B/WG7B/LG8B	AVG	91.36	97.26	82.97	63.33	71.83	67.69
GG2B/SG2B/WG7B	OR	90.96	92.98	70.93	81.33	75.78	70.50
GG2B/SG2B/WG7B	AND	87.25	99.58	93.48	28.67	43.88	47.65
GG2B/SG2B/WG7B	AVG	92.12	98.25	88.32	63.00	73.54	70.42
GG2B/SG2B/LG8B	OR	90.61	92.49	69.60	81.67	75.15	69.75
GG2B/SG2B/LG8B	AND	87.01	99.72	95.24	26.67	41.67	46.47
GG2B/SG2B/LG8B	AVG	91.30	98.04	86.41	59.33	70.36	67.05
GG2B/WG7B/LG8B	OR	90.90	92.77	70.49	82.00	75.81	70.55
GG2B/WG7B/LG8B	AND	90.67	99.65	96.64	48.00	64.14	64.29
GG2B/WG7B/LG8B	AVG	**92.70**	97.47	85.37	70.00	76.92	73.14
SG2B/WG7B/LG8B	OR	92.17	95.44	77.97	76.67	77.31	72.59
SG2B/WG7B/LG8B	AND	87.25	**100.00**	**100.00**	26.67	42.11	48.06

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Table 112 – continued from previous page

Model	Logic	Accuracy(%)	Specificity(%)	Precision(%)	Recall(%)	F1Score(%)	MCC(%)
SG2B/WG7B/LG8B	AVG	90.72	98.25	86.84	55.00	67.35	64.46
LG1B/GG2B/SG2B/WG7B	OR	49.33	40.42	24.47	91.67	38.62	25.53
LG1B/GG2B/SG2B/WG7B	AND	85.10	99.72	92.16	15.67	26.78	34.43
LG1B/GG2B/SG2B/WG7B	AVG	91.88	95.58	77.97	74.33	76.11	71.25
LG1B/GG2B/SG2B/LG8B	OR	49.28	40.28	24.49	92.00	38.68	25.71
LG1B/GG2B/SG2B/LG8B	AND	84.99	99.72	91.84	15.00	25.79	33.58
LG1B/GG2B/SG2B/LG8B	AVG	90.61	95.02	74.64	69.67	72.07	66.49
LG1B/GG2B/WG7B/LG8B	OR	49.39	40.49	24.49	91.67	38.65	25.57
LG1B/GG2B/WG7B/LG8B	AND	86.84	99.72	95.06	25.67	40.42	45.49
LG1B/GG2B/WG7B/LG8B	AVG	92.06	95.09	76.90	77.67	77.28	72.47
LG1B/SG2B/WG7B/LG8B	OR	50.03	41.68	24.45	89.67	38.43	24.72
LG1B/SG2B/WG7B/LG8B	AND	85.16	**100.00**	**100.00**	14.67	25.58	35.26
LG1B/SG2B/WG7B/LG8B	AVG	91.36	96.77	81.07	65.67	72.56	68.03
GG2B/SG2B/WG7B/LG8B	OR	90.67	92.21	69.25	83.33	75.64	70.39
GG2B/SG2B/WG7B/LG8B	AND	87.13	**100.00**	**100.00**	26.00	41.27	47.43
GG2B/SG2B/WG7B/LG8B	AVG	92.64	97.05	83.66	71.67	77.20	73.15
LG1B/GG2B/SG2B/WG7B/LG8B	OR	49.22	40.14	24.51	**92.33**	38.74	25.89
LG1B/GG2B/SG2B/WG7B/LG8B	AND	85.10	**100.00**	**100.00**	14.33	25.07	34.85
LG1B/GG2B/SG2B/WG7B/LG8B	AVG	91.54	97.96	86.32	61.00	71.48	68.07